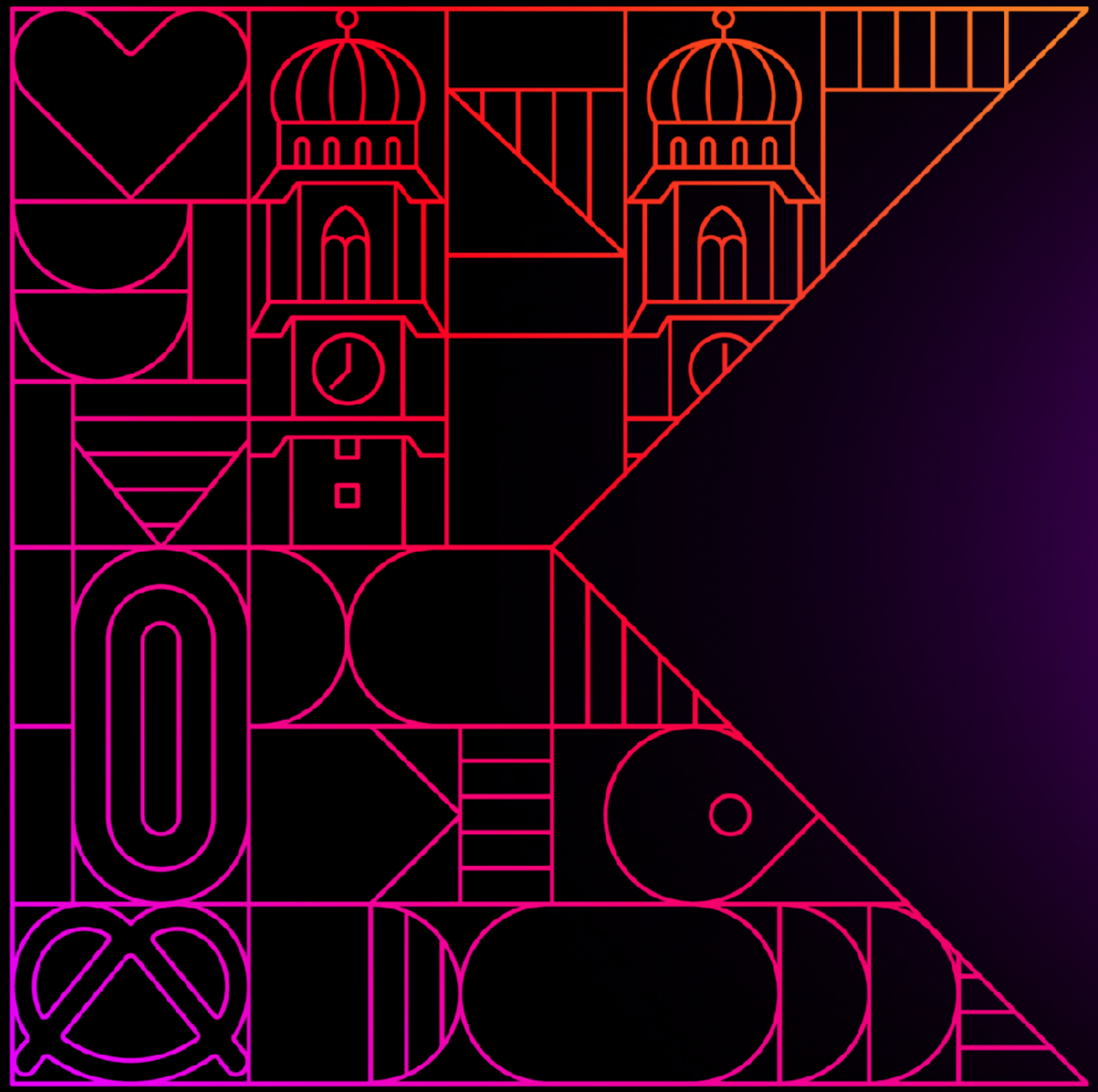
KotlinConf 2026

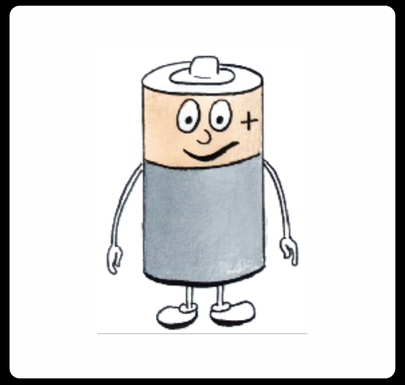
Redefining Machine Learning with Kotlin

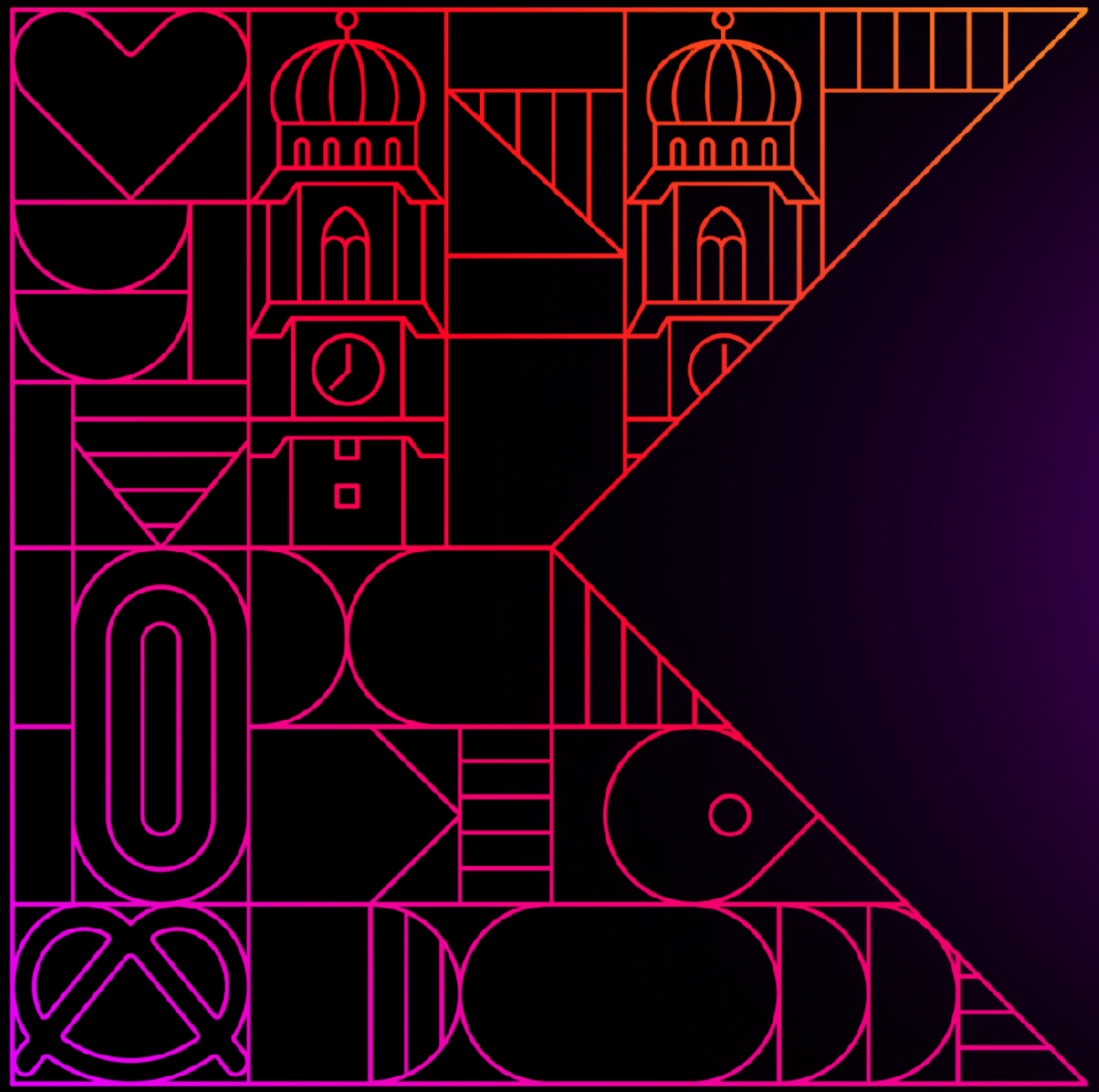


Michal Harakal
Deutsche Telekom AG

KotlinConf 2026

Redefining Machine Learning with Kotlin





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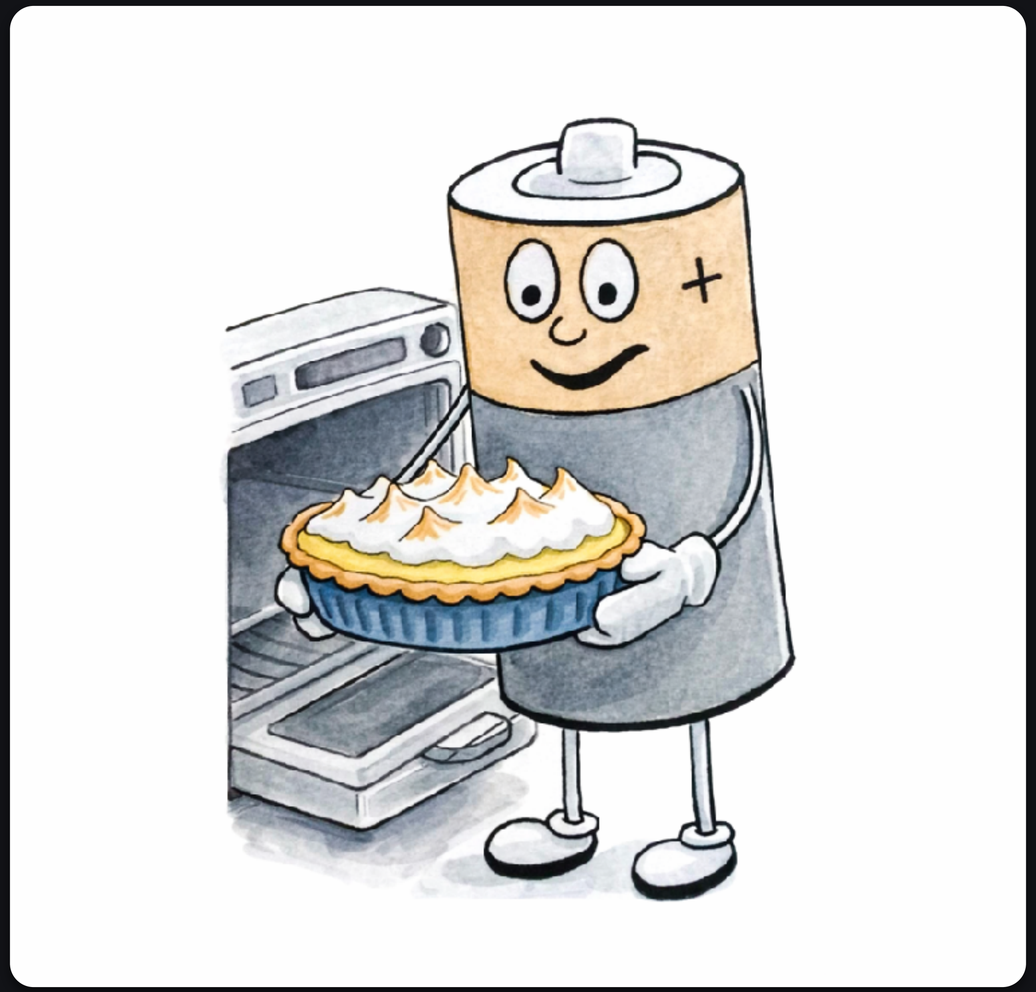
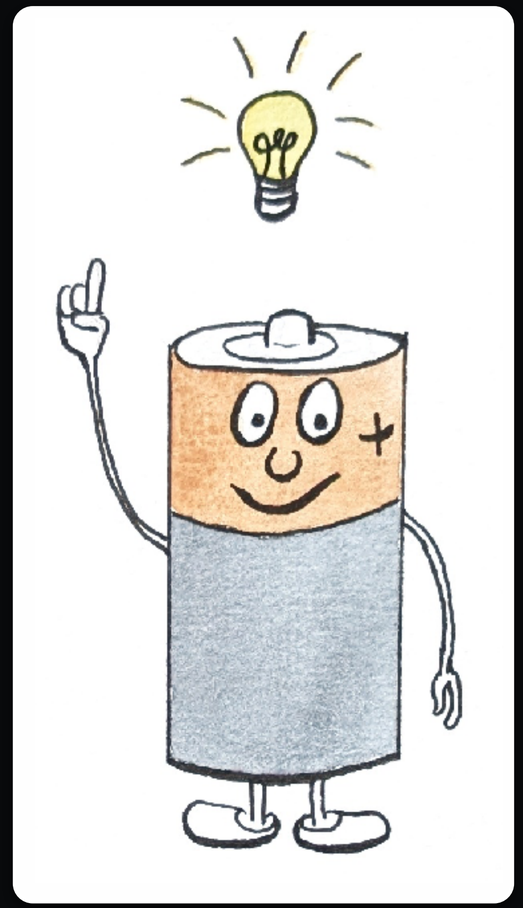


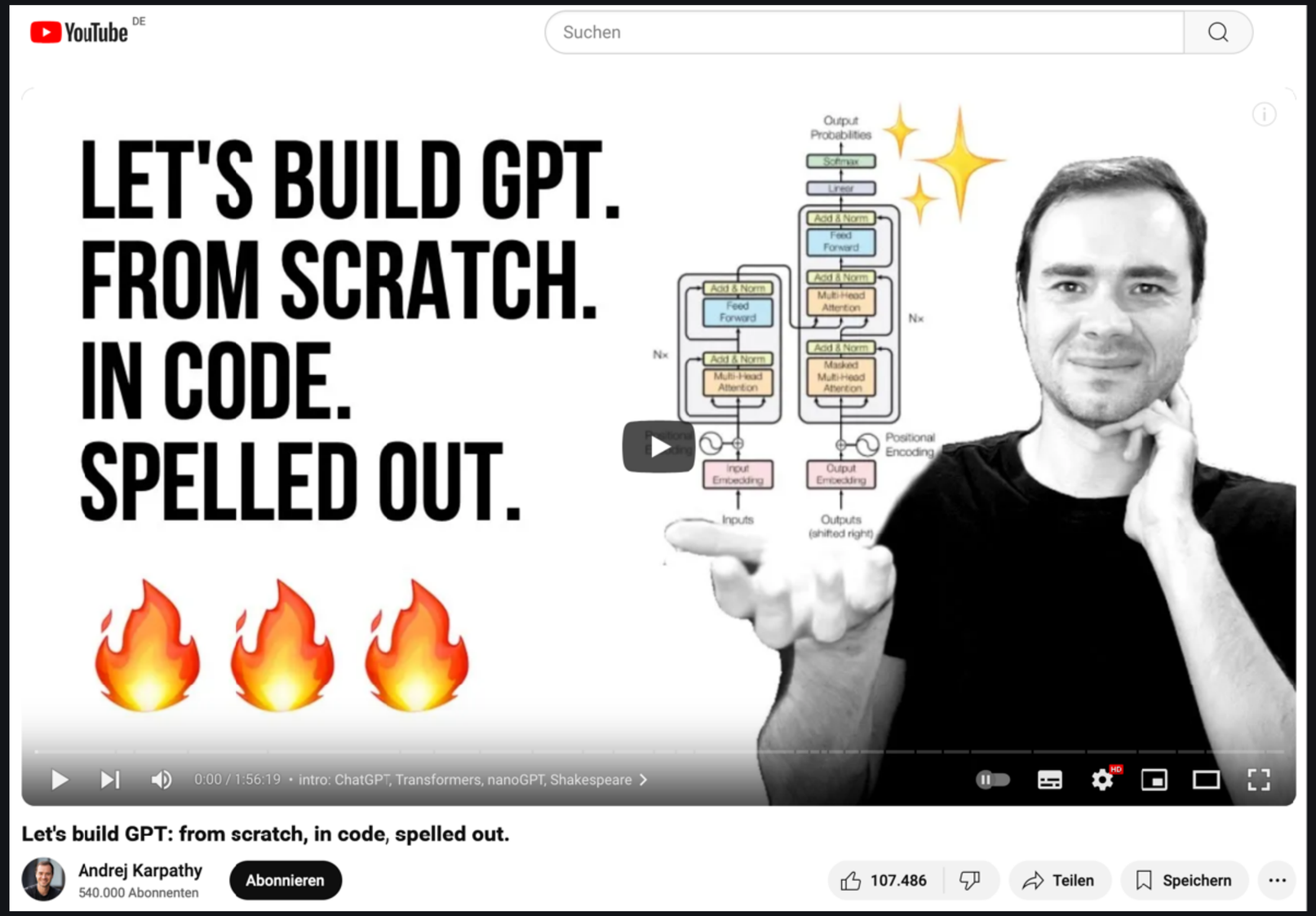
AI generated images based on Bateryman by (C) Adriana Harakalova, 2024

What I cannot create,
I do not understand.
Know how to solve every problem that has been solved.

Richard Feynman, 1988



(C) Adriana Harakalova, 2024



https://www.youtube.com/watch?v=kCc8FmEb1nY

Let's build GPT: from scratch, in code, spelled out.

Andrej Karpathy

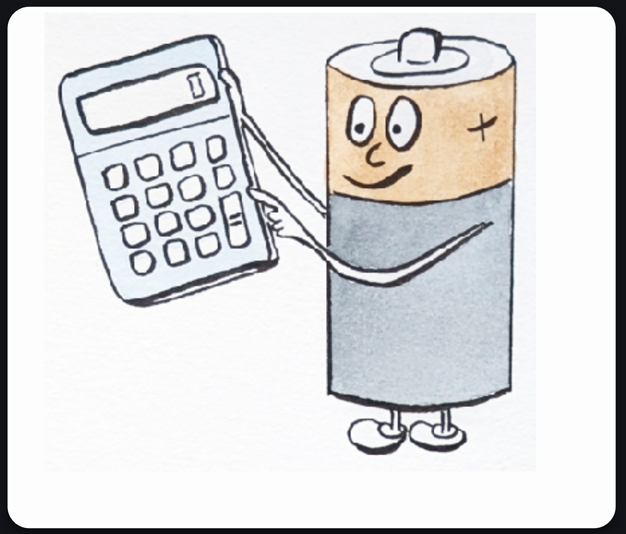
- GTP-2 level model with about 10 million parameters
- 250 lines of code

Let's build GPT: from scratch, in code, spelled out.

Andrej Karpathy

- GTP-2 level model with about 10 million parameters
- 250 lines of code
- written in PyTorch

Transformers 101

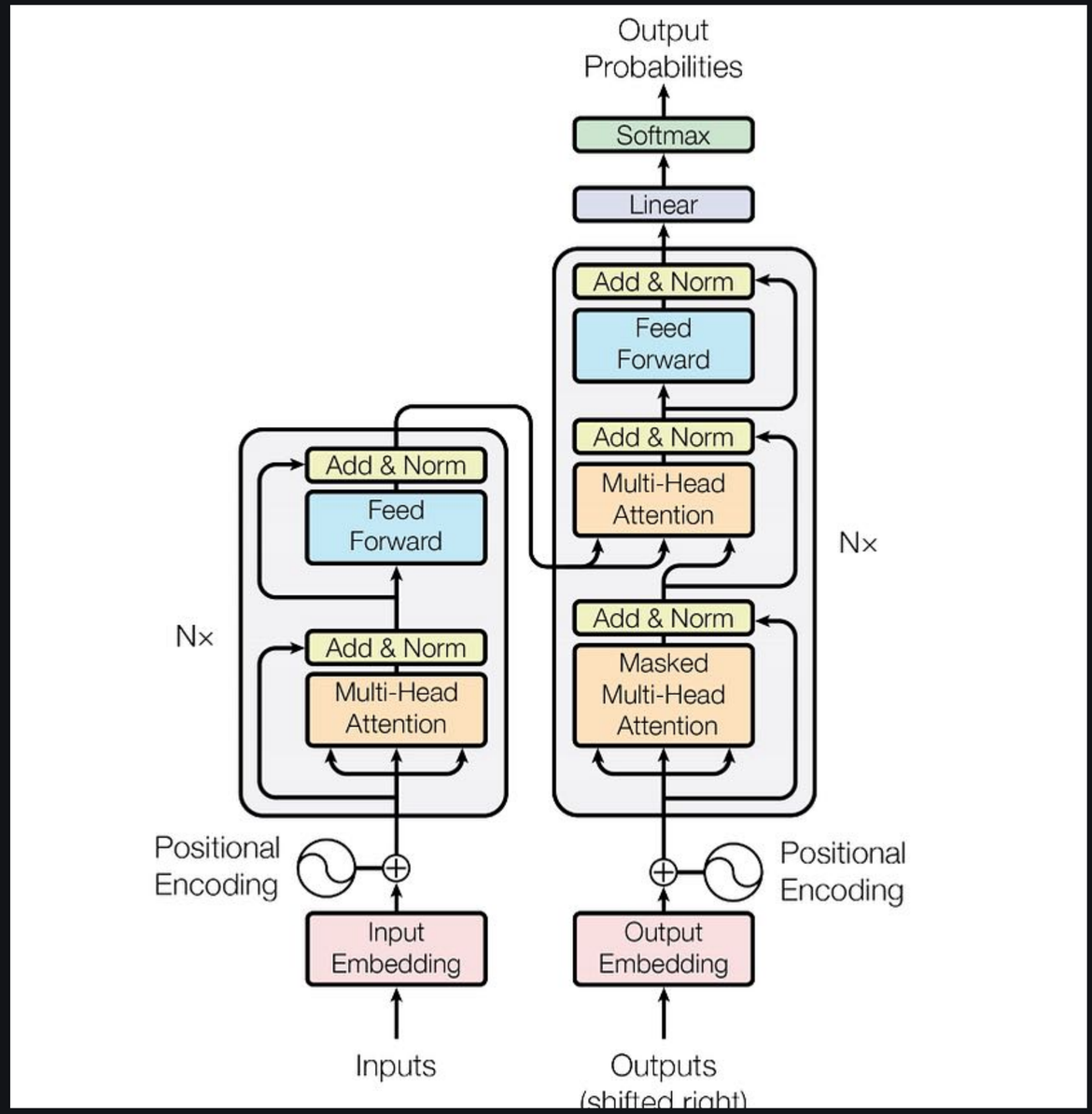


(C) Adriana Harakalova, 2024

Attention Is All You Need

Google Brain, Google Research et al.

- Published 2017
- Transformer model architecture
- Efficient computation as goal
- Encoder/Decoder for Translations



Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin

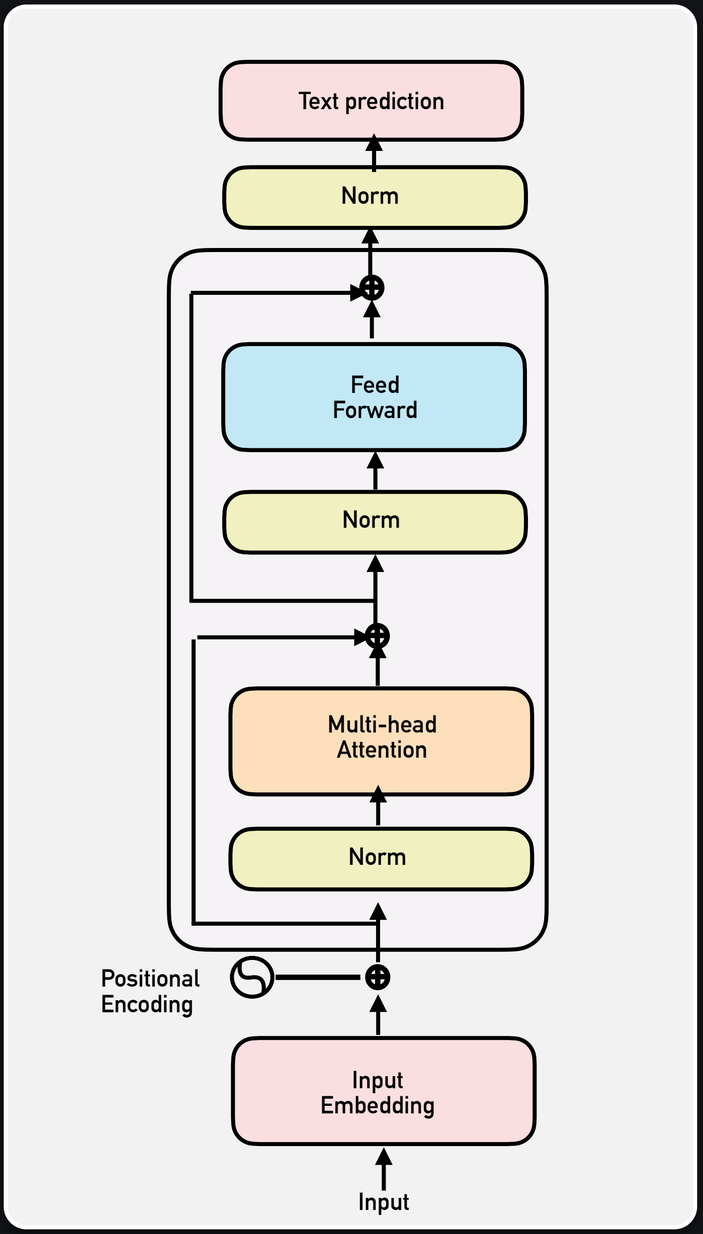
Decoder only

GTP-2

- 117M parameters
- 12 decoder layers
- 768 embedding

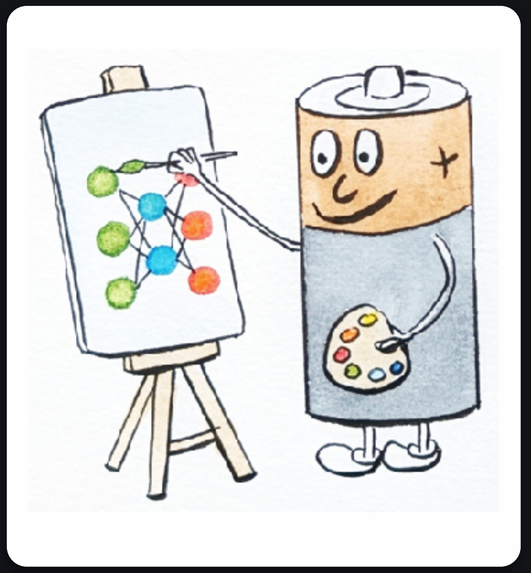
A. Karpathy's model

- 10M parameters
- 12 layers, 12 heads
- 1024 context



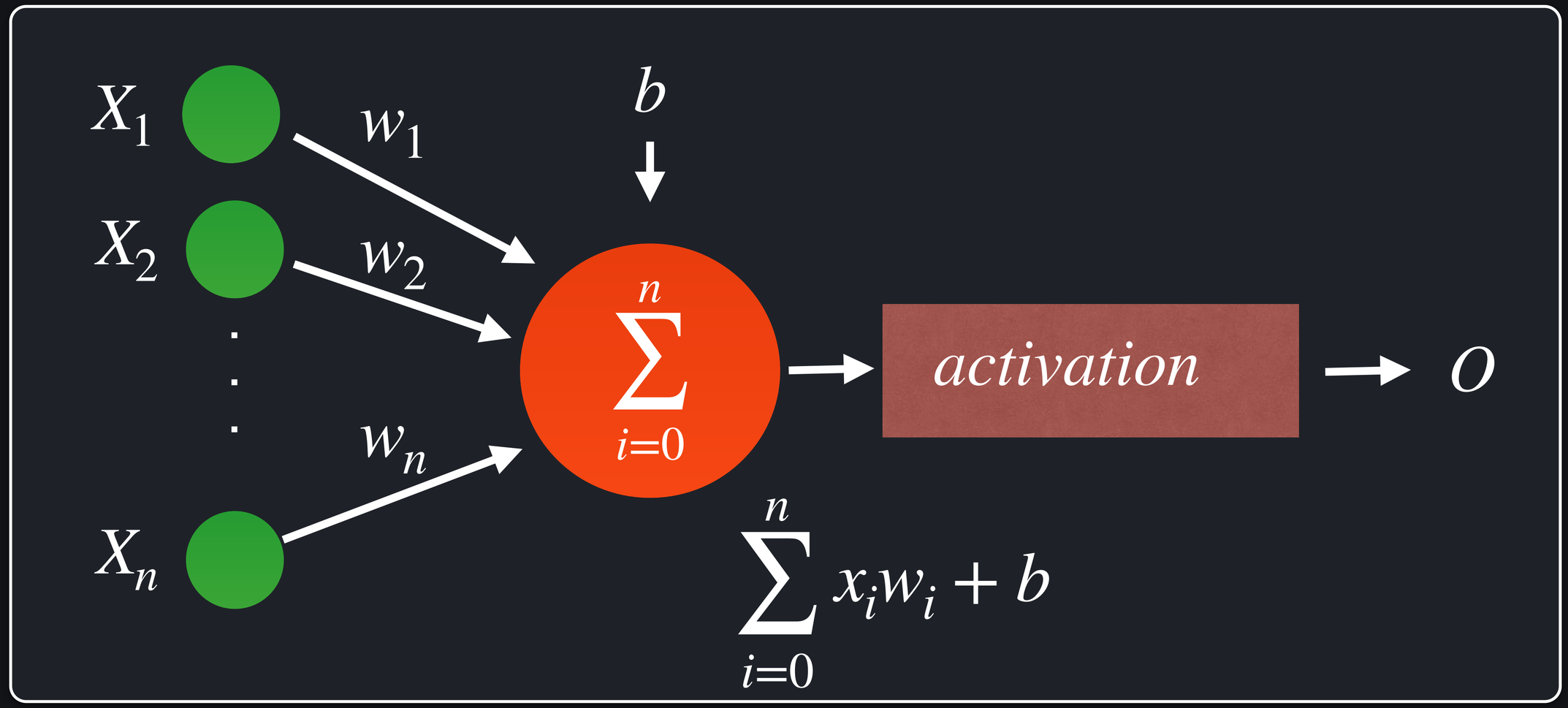
Deep neural networks

Feed
Forward

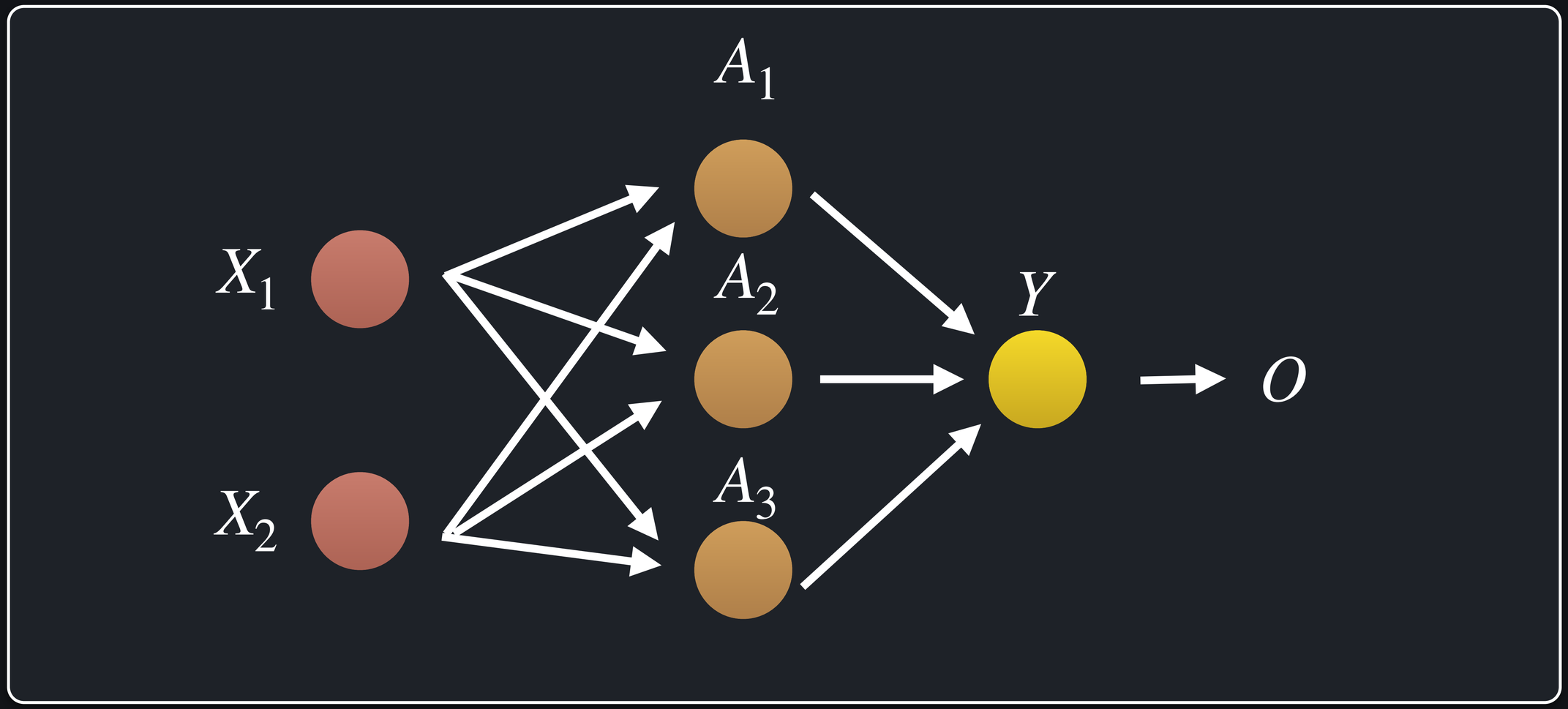


(C) Adriana Harakalova, 2024

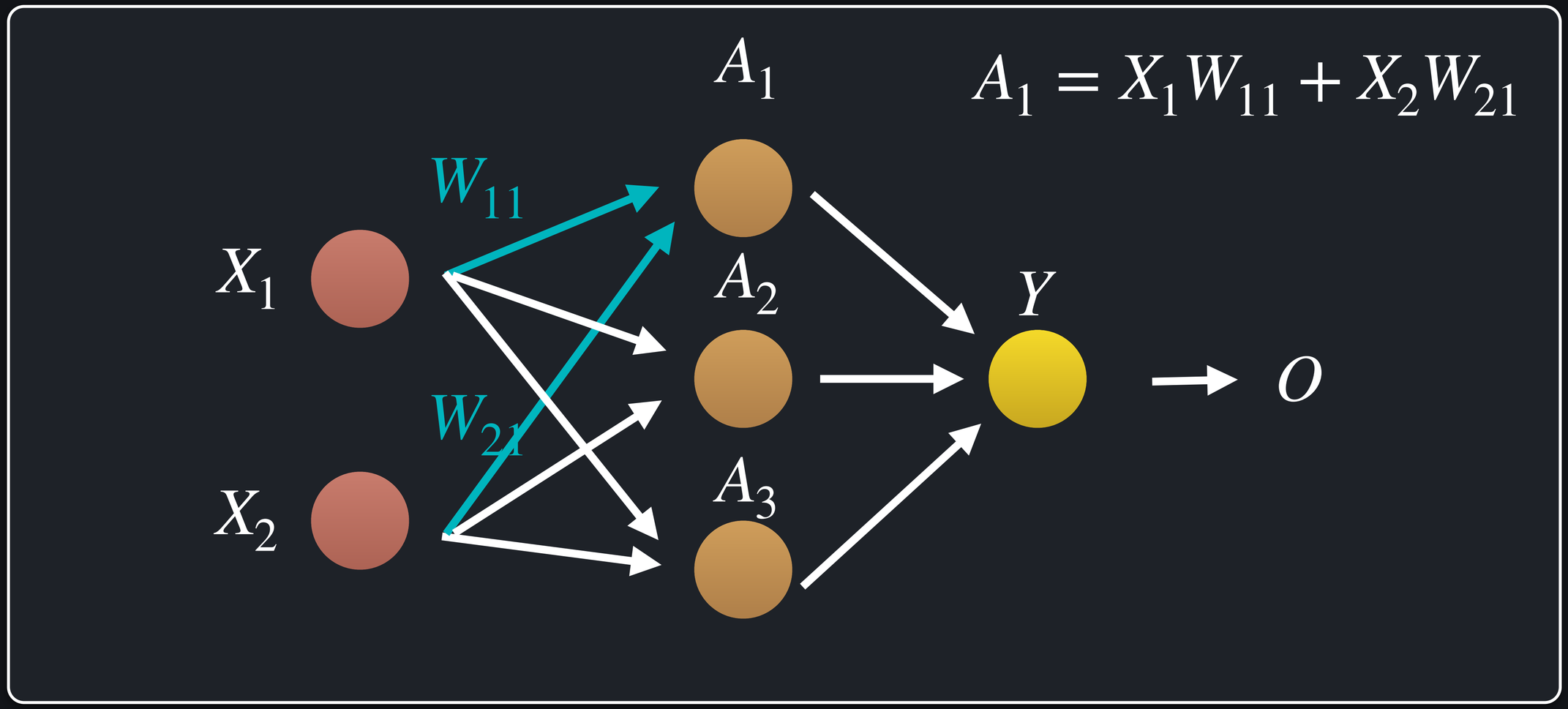
Neuron



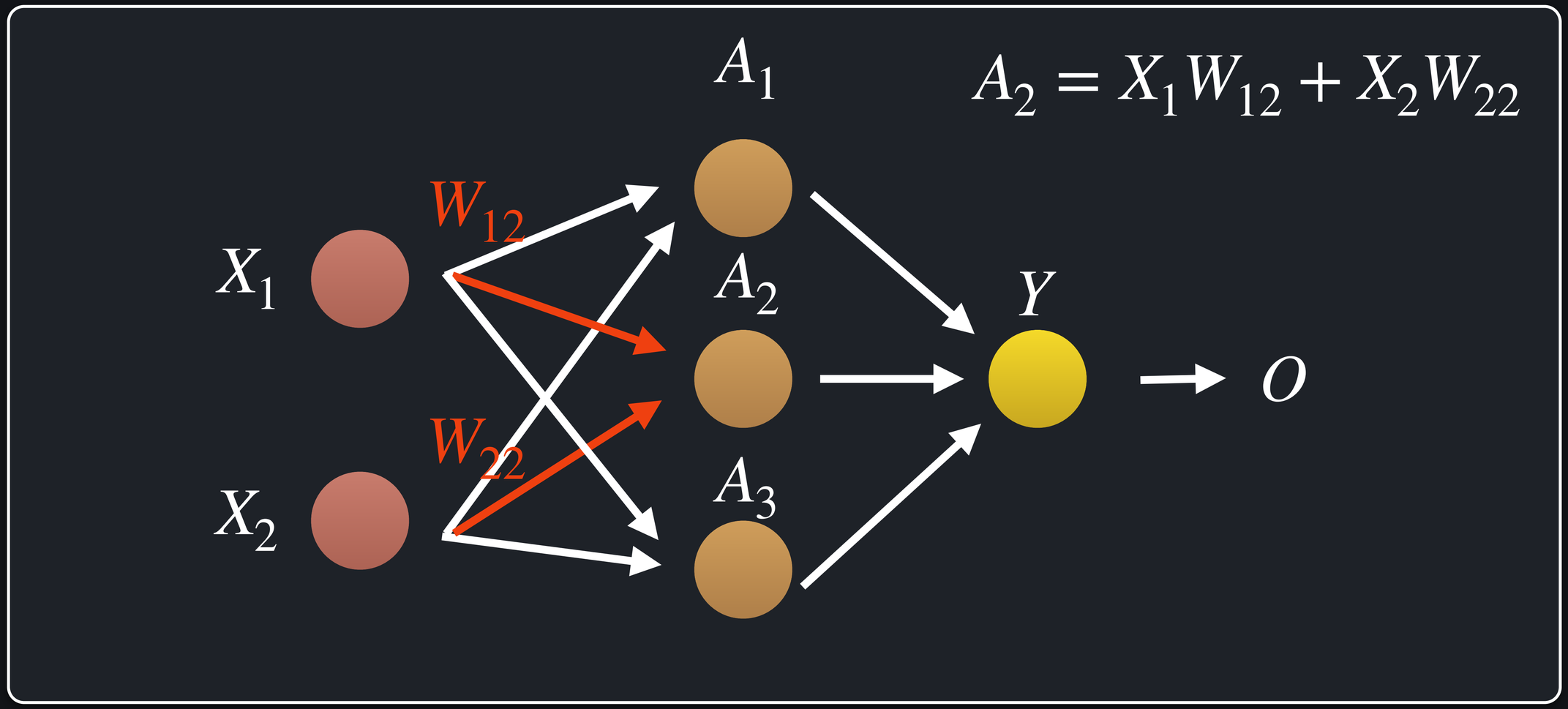
MLP - Multilayer Perceptron



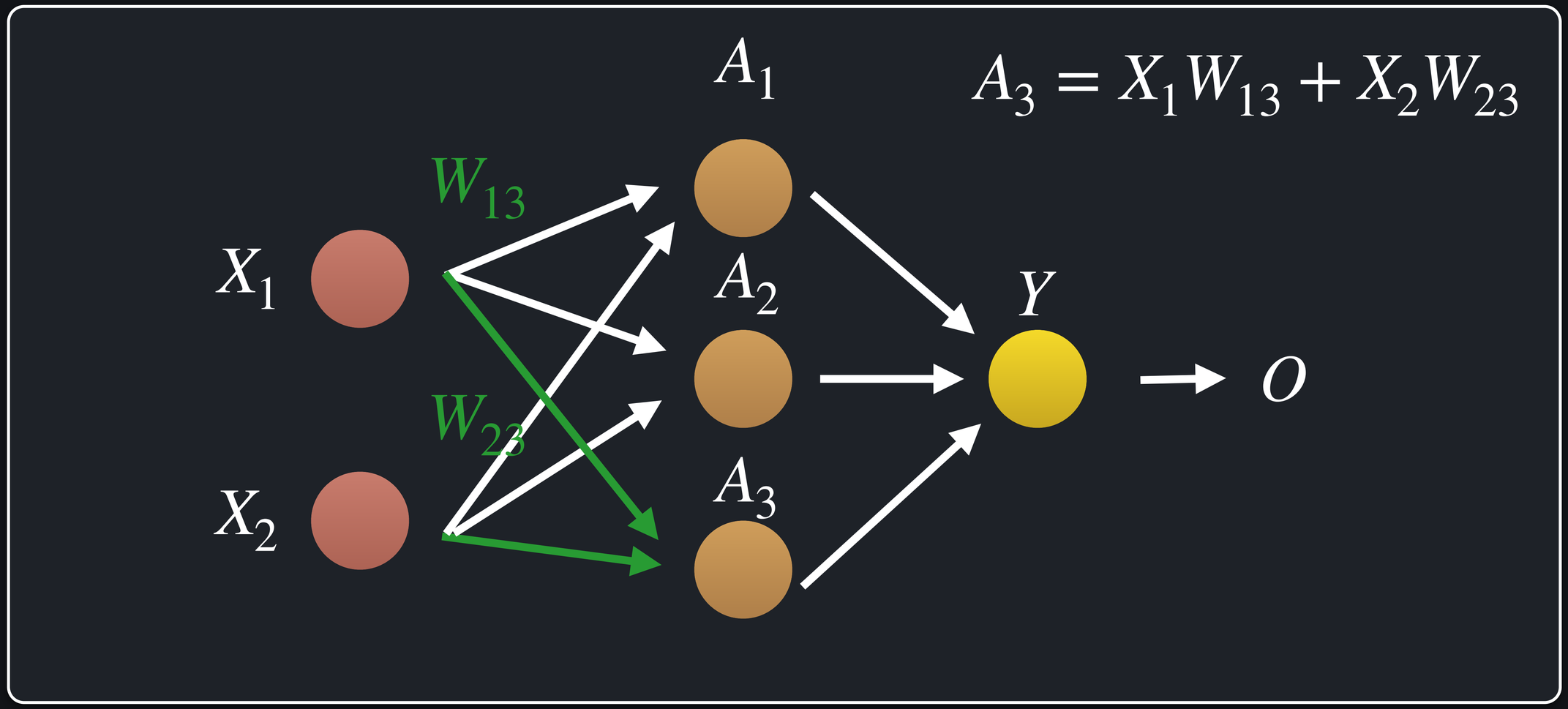
Forward propagation



Forward propagation



Forward propagation



Matrix Multiplication

$$A_{11} = X_1 W_{11} + X_2 W_{21}$$

$$A_{12} = X_1 W_{12} + X_2 W_{22}$$

$$A_{13} = X_1 W_{13} + X_2 W_{23}$$

$$A = X \cdot W$$

$$W = \begin{bmatrix} W_{11} & W_{12} & W_{13} \\ W_{21} & W_{22} & W_{23} \end{bmatrix}$$

$$X = \begin{bmatrix} X_1 & X_2 \end{bmatrix}$$

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{12} \end{bmatrix}$$

$$A = X \cdot W$$

Matrices multiplication

$$A = X \cdot W$$

In Kotlin with lists of doubles

```kotlin
class Matrix(private val data: List<List<Double>>) {
    private val numRows = data.size
    private val numCols = data.firstOrNull()?.size ?: 0

    fun matmul(other: Matrix): Matrix {
        val resultData = List(numRows) { i ->
            List(other.numCols) { j ->
                (0 until numCols).sumOf { k ->
                    data[i][k] * other.data[k][j]
                }
            }
        }
        return Matrix(resultData)
    }
}
```

Faster matrices multiplication

$$A = X \cdot W$$

Support for SIMD - DoubleArray

```
class Matrix(
    val rows: Int,
    val cols: Int,
    val data: DoubleArray   // row-major, flat
) {
    init {
        require(data.size == rows * cols)
    }

    constructor(nested: List<List<Double>>) : this(
        nested.size,
        nested.firstOrNull()?.size ?: 0,
        DoubleArray(nested.size * (nested.firstOrNull()?.size ?: 0)).also { flat
->
            val c = nested.firstOrNull()?.size ?: 0
            for (i in nested.indices) {
                val row = nested[i]
                require(row.size == c)
                for (j in 0 until c) flat[i * c + j] = row[j]
            }
        }
    )
```

Faster matrices multiplication

$$A = X \cdot W$$

Support for SIMD - Loop reorder and manual hoisting

```
fun fasterMatmul(other: Matrix): Matrix {
   val m = rows val; k = cols
    val n = other.cols; val a = this.data
    val b = other.data; val c = DoubleArray(m * n)   // zero-initialized
    var i = 0
    while (i < m) {
        val iC = i * n
        val iA = i * k
        var kk = 0
        while (kk < k) {
            val aik = a[iA + kk]           // hoisted scalar
            val bRow = kk * n              // hoisted base index
            if (aik != 0.0) {              // optional: skip zeros (sparse-friendly)
                var j = 0
                while (j < n) {
                    c[iC + j] += aik * b[bRow + j]
                    j++
                }
            }
            kk++
        }
        i++
    }
```

8x Faster matrices multiplication $A = X \cdot W$

Support for SIMD - Loop reorder and manual hoisting

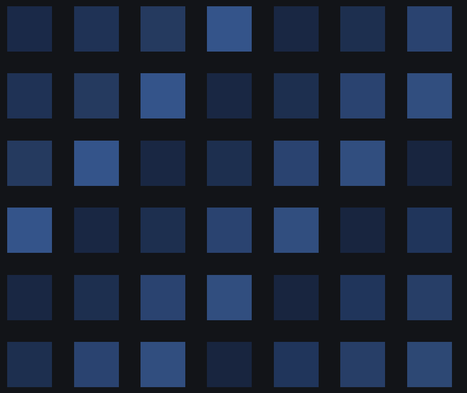
```
fun fasterMatmul(other: Matrix): Matrix {
   val m = rows val; k = cols
    val n = other.cols; val a = this.data
    val b = other.data; val c = DoubleArray(m *
    var i = 0
    while (i < m) {
        val iC = i * n
        val iA = i * k
        var kk = 0
        while (kk < k) {
            val aik = a[iA + kk]           // ho
            val bRow = kk * n              // ho
            if (aik != 0.0) {              // op
                var j = 0
                while (j < n) {
                    c[iC + j] += aik * b[bRow + j
                    j++
                }
            }
            kk++
        }
        i++
    }
```

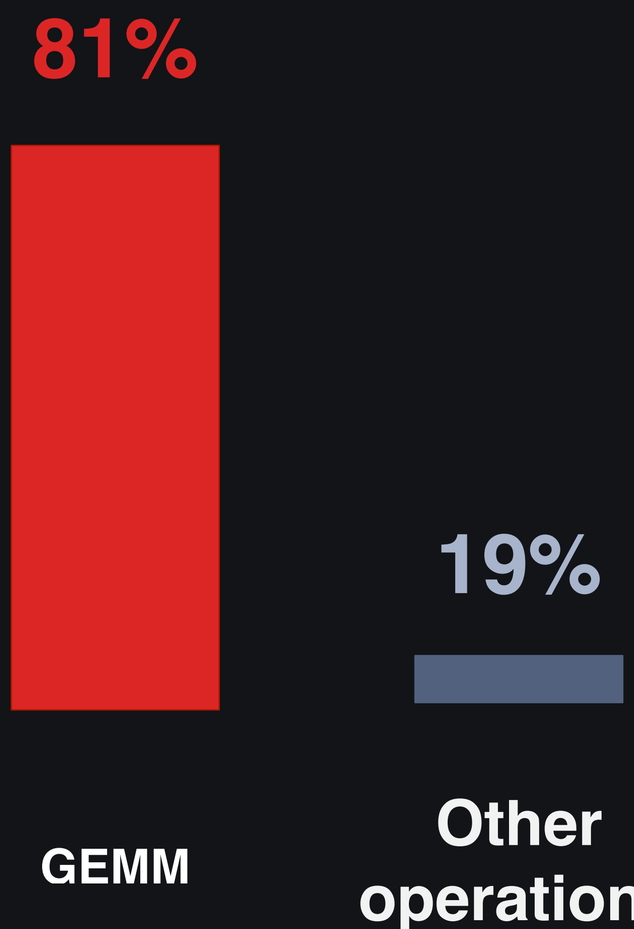
Premature Optimization ???

81% of the compute in GTP-2 XL

is general matrix multiplication

$$A = X \cdot W$$





Why it matters

Every speedup in matrix kernels compounds across training and inference.

https://arxiv.org/html/2404.11788v4

ReLU

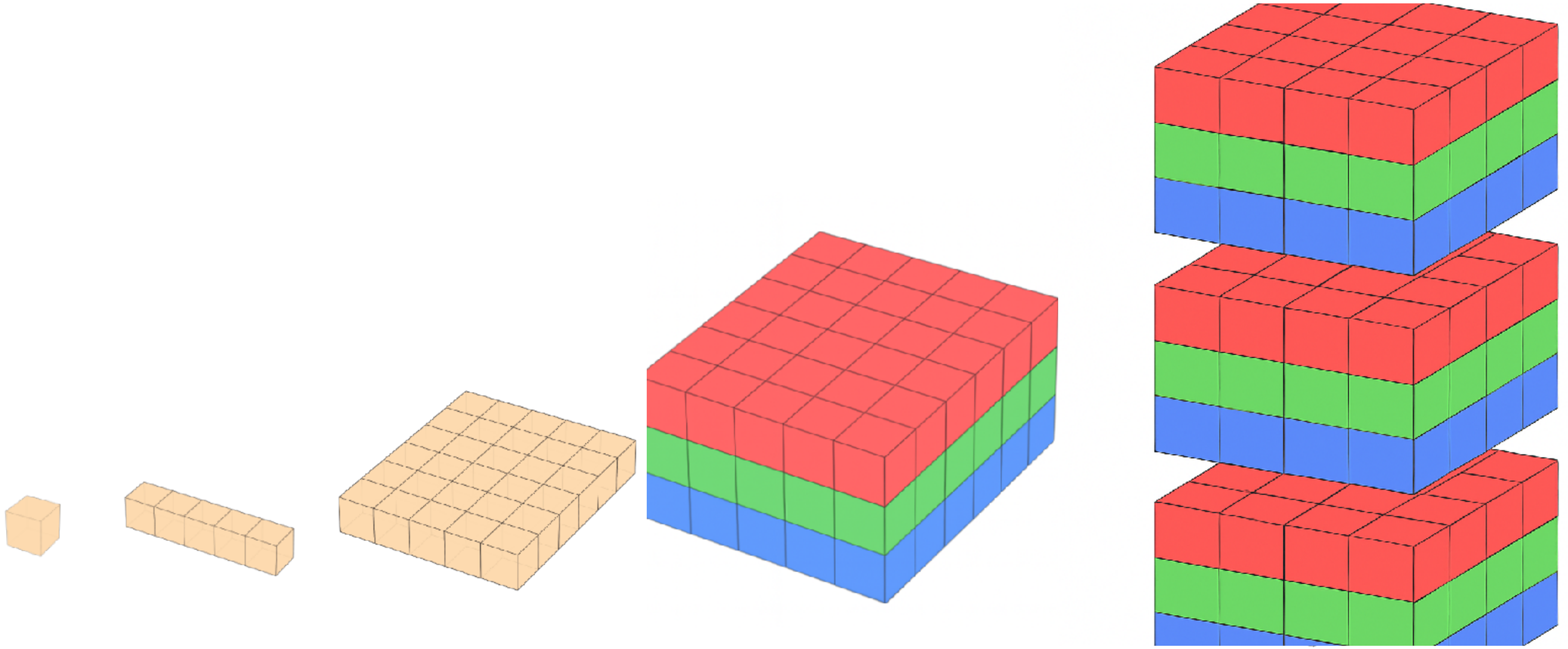
Activation function

$$\text{ReLU}(x) = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$

```
fun Matrix.relu(): Matrix {
    val src = this.data
    val dst = DoubleArray(src.size)
    for (i in src.indices) {
        dst[i] = if (src[i] > 0.0) src[i] else 0.0
    }
    return Matrix(rows, cols, dst)
}
```

ML/AI Data structures

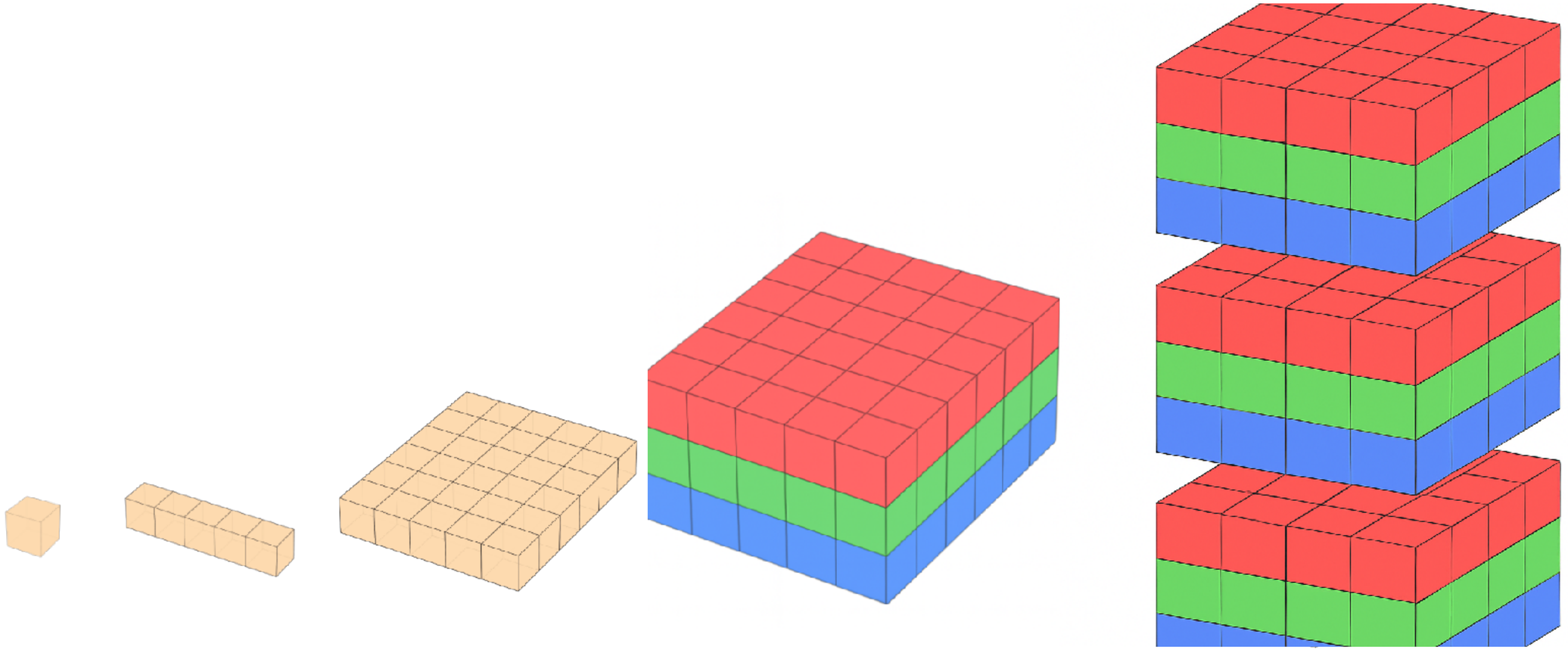
Scalar



Scalar Vector Matrix Tensor Batch

ML/AI Data structures

All tensors with a rank

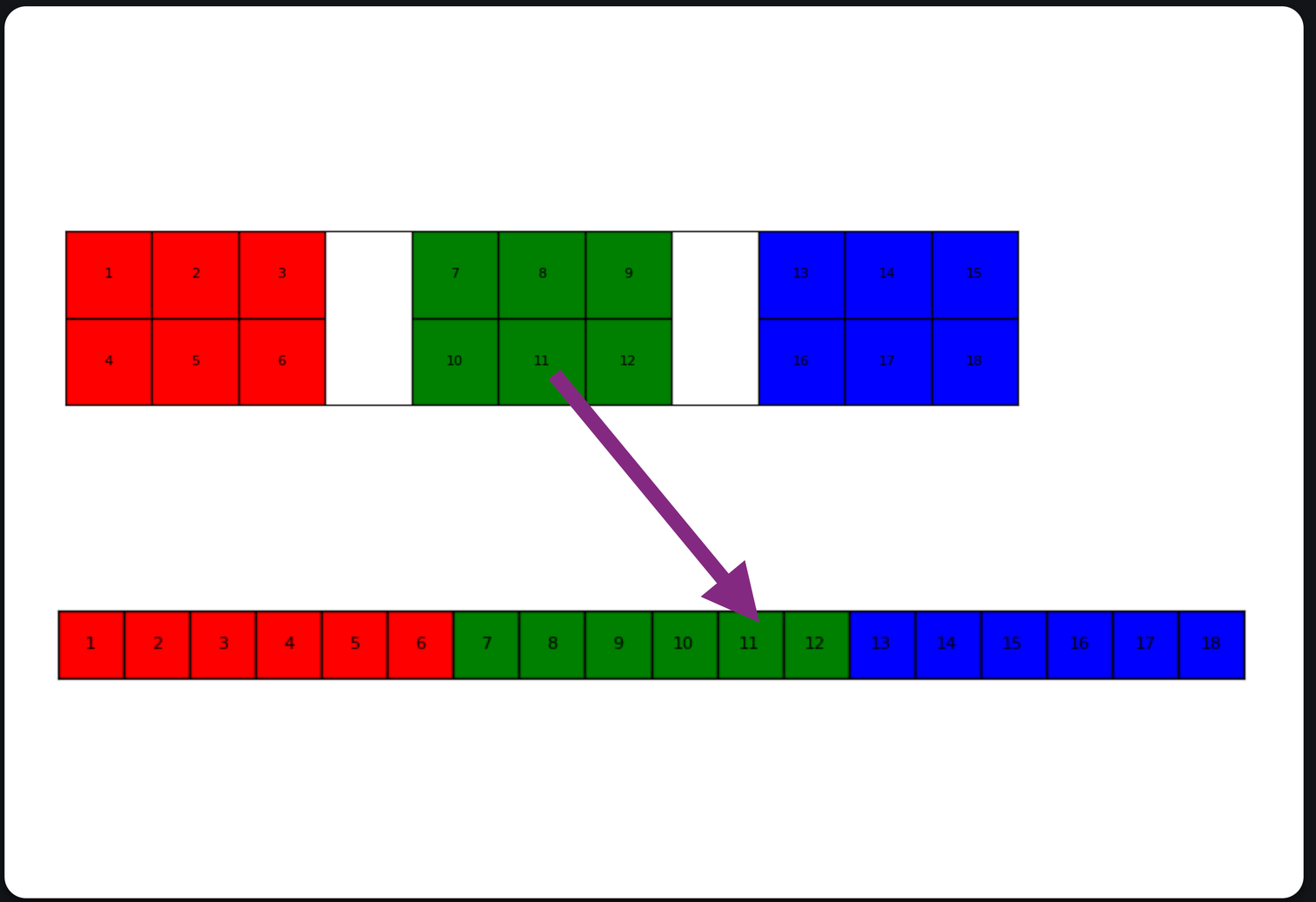


Scalar	Vector	Matrix	Tensor	Tensor
Rank=0	Rank=1	Rank=2	Rank=3	Rank=4

Memory layout

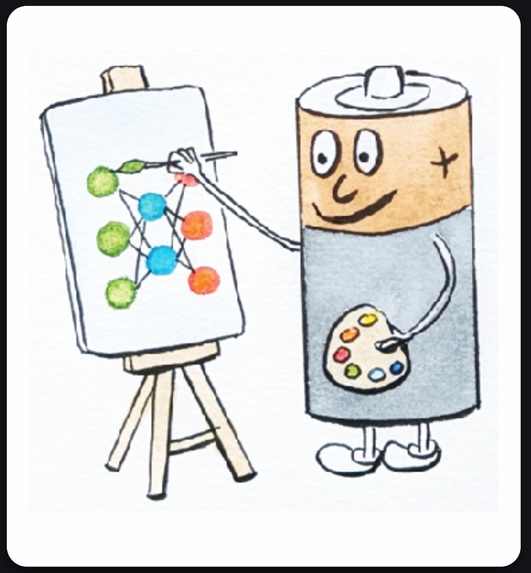
Tensors

- Elements order - **Row major**
- convention
  - BCWH, BWHC -picture
  - B, T, C - Transformers

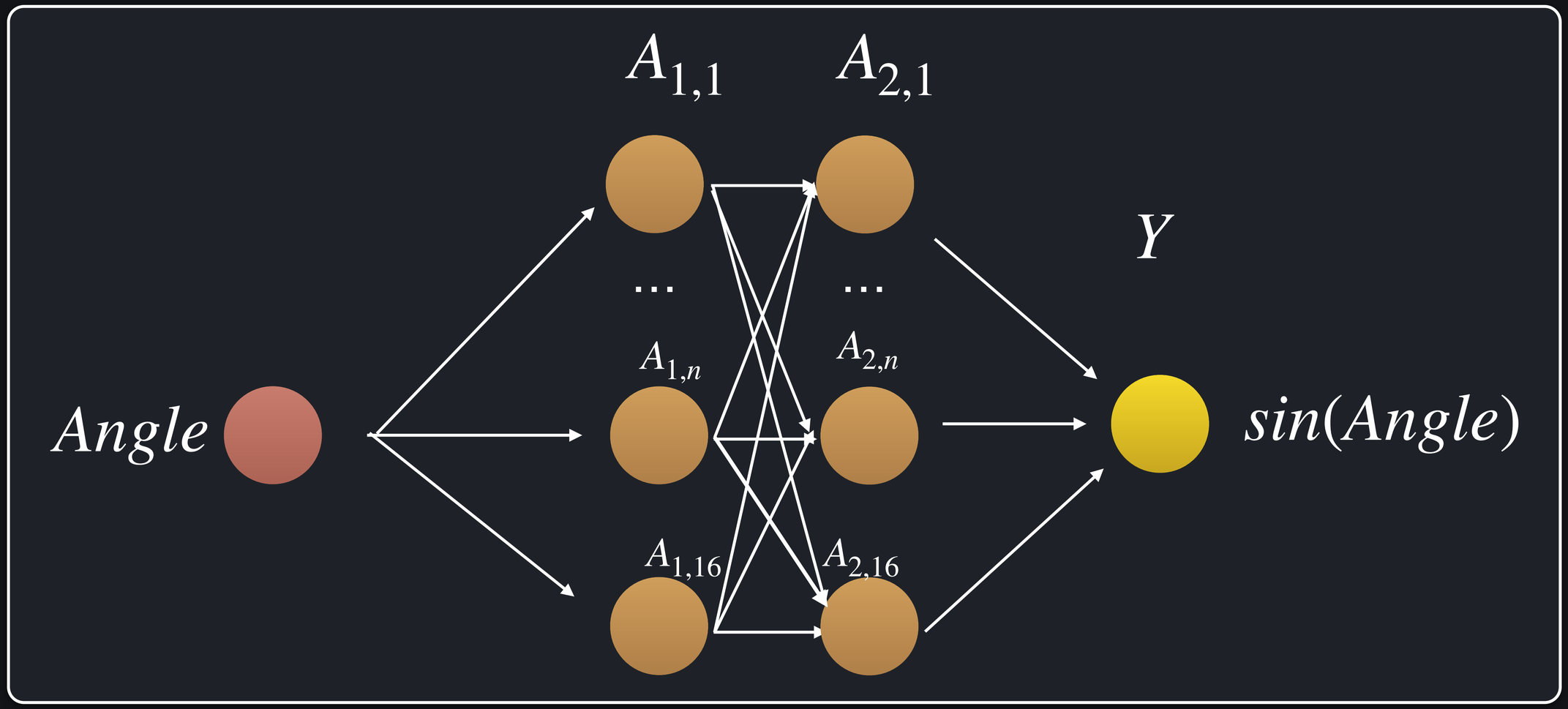


Simple deep neural network example

Feed
Forward



Sinus approximation with neuronal network



Forward Propagation in Kotlin

Hidden layer

```
class Linear() {
    val weights:Tensor
    val bias:Tensor

     fun forward(input: Tensor): Tensor {
       val output = input.matmul(weight.value.t()) + bias.value
       return output
     }
}
```

Forward Propagation in Kotlin

MLP

```
class SineMLP : Module() {
    private val inputLayer: Module = Linear(1, 16)
    private val hiddenLayer1: Module = Linear(16, 16)
    private val hiddenLayer2: Module = Linear(16, 16)
    private val outputLayer: Module = Linear(16, 1)
    private val relu: Module = ReLU()

    override fun forward(input: Tensor): Tensor {
        var x = input
        x = inputLayer.forward(x)
        x = relu.forward(x)
        x = hiddenLayer1.forward(x)
        x = relu.forward(x)
        x = hiddenLayer2.forward(x)
        x = relu.forward(x)
        x = outputLayer.forward(x)
        return x
    }
}
```

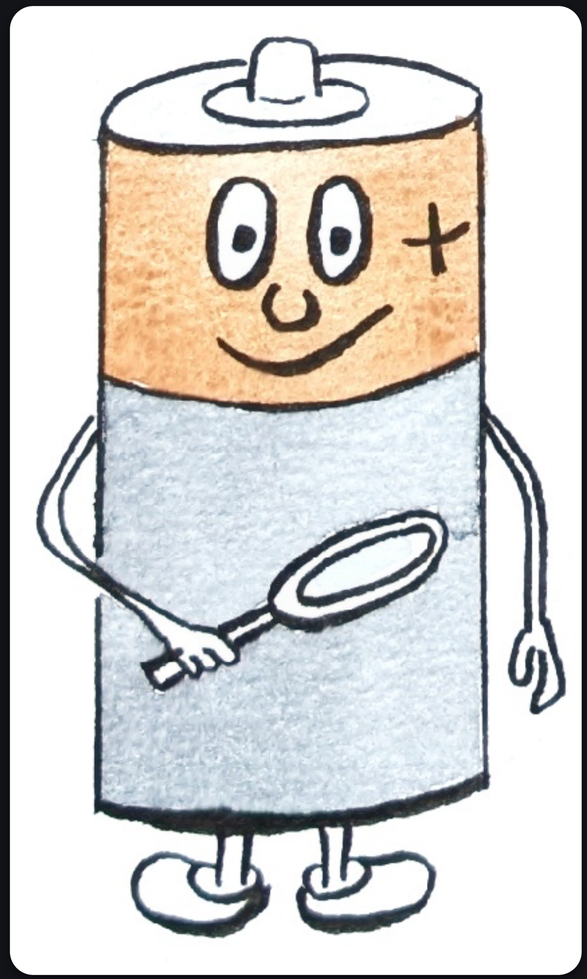
Forward Propagation in Kotlin

MLP with DSL

```
class SineNN(override val name: String="sin") : Module() {

    private val sineModule =  sequential{
        input(1)
        linear(16, "layer1") {
            activation = relu
        }
        linear(16, "layer2") {
            activation = relu
        }
        linear(1, "output_layer")
    }
    override fun forward(input: Tensor): Tensor =
        sineModule.forward(input)
}
```

Reality check



ML Babylon

Many languages, many platforms, many technologies

Languages

- **Halide**
  - Focus on Pictures and Tensors
- **Julia**
  - Dynamically types
  - Flux.jl Lux.jl for **ML**
  - popular in academia

HW/Plattformen

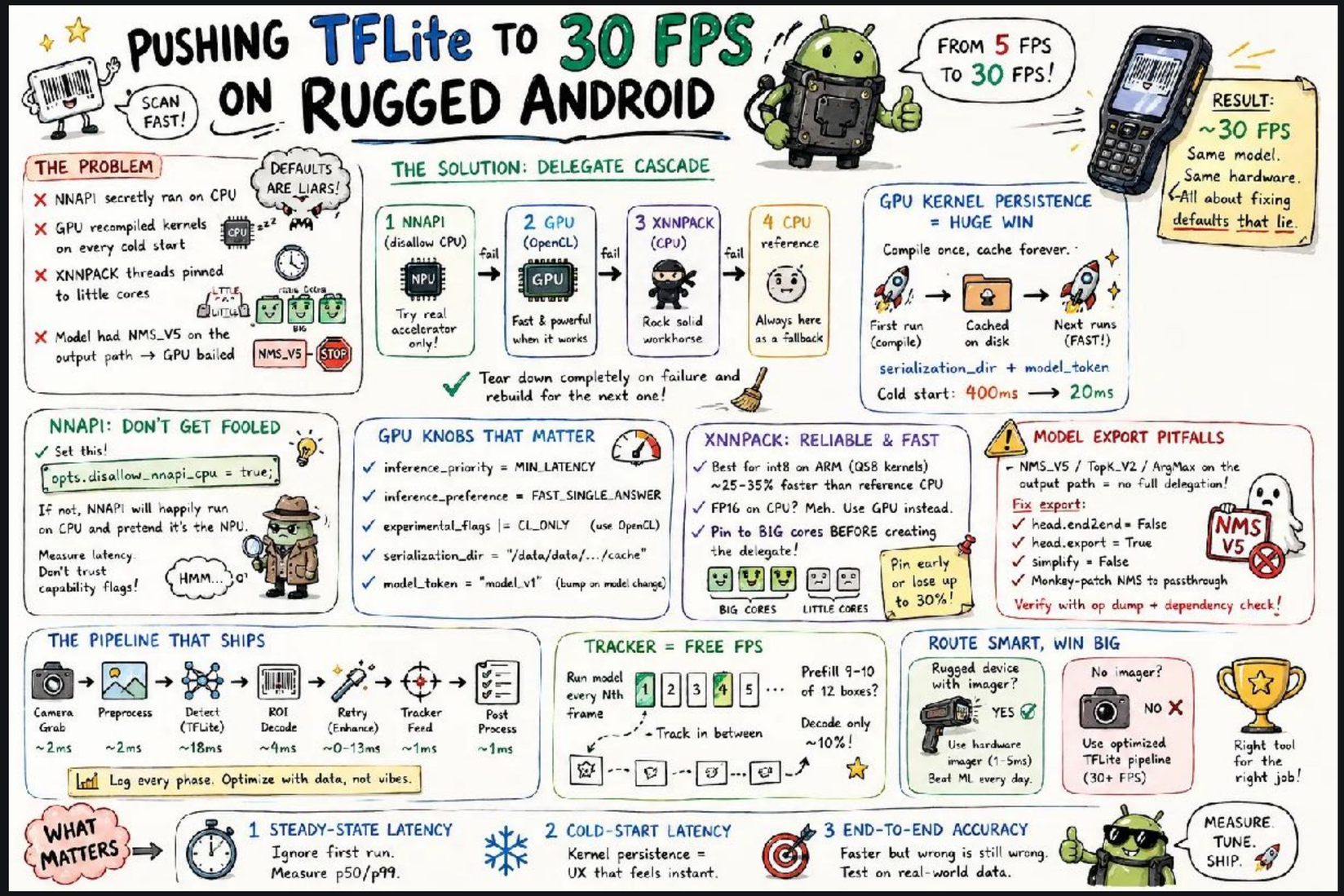
- **CUDA**
  - **Nvidia**
- **Metal**
  - Apple
- **Vulcan, NPU**
  - Android

Focus on AI

- Mojo
- Pythonic
- MLIR / LLVM IR
- ONNX

From 5 fps to 30 fps: What the TFLite Defaults Are Hiding

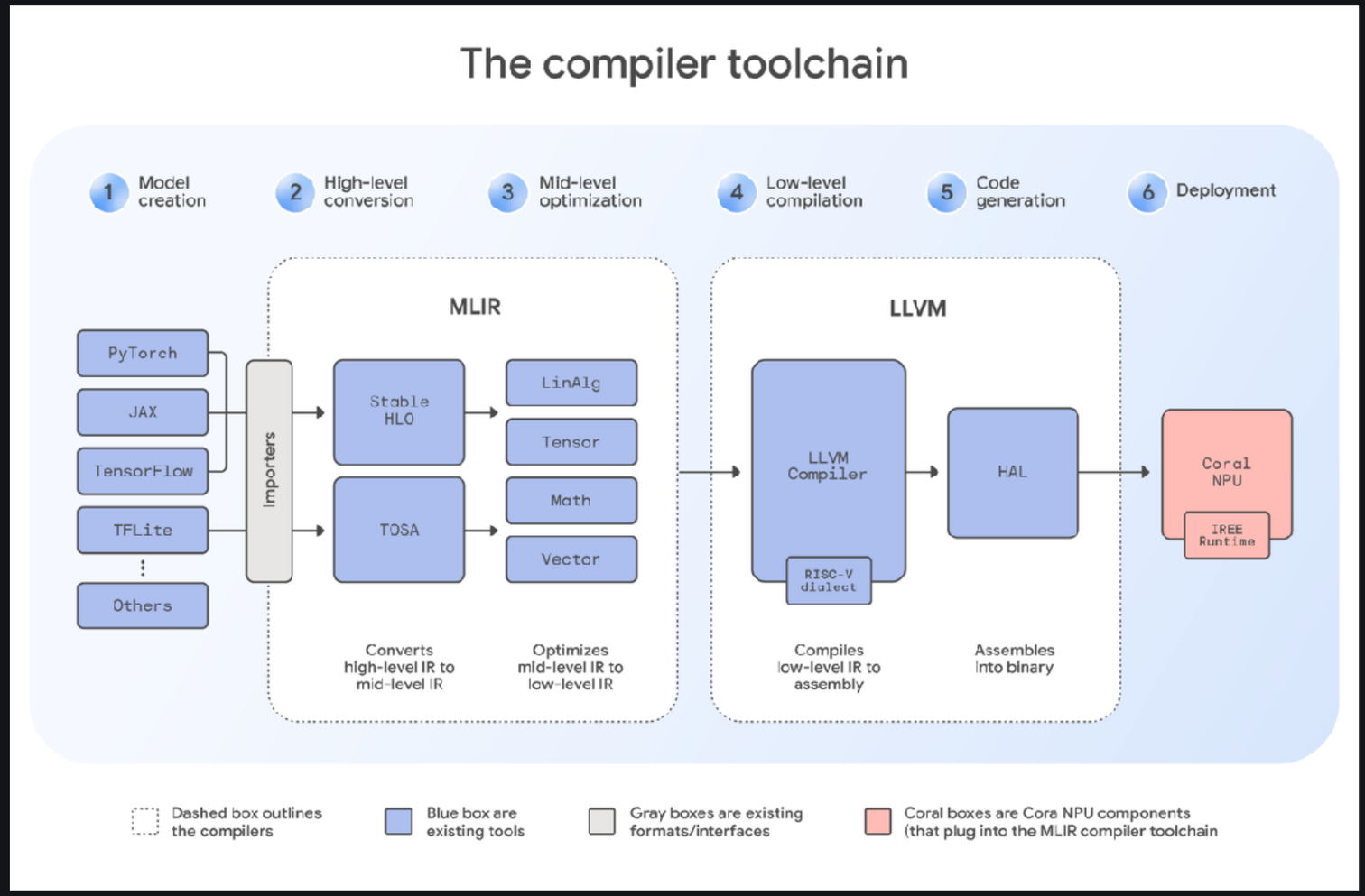
Kashif Mehmood



https://proandroiddev.com/from-5-fps-to-30-fps-what-the-tflite-defaults-are-hiding-a8fbf0d20ff6

Google - Coral NPU

Compiler based approach, November 2025

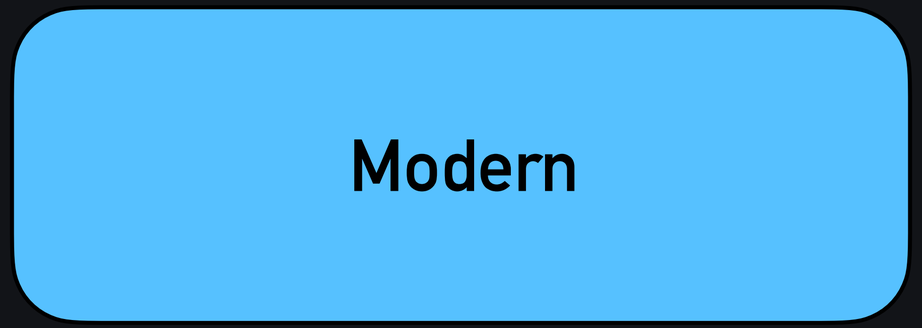


https://developers.googleblog.com/en/introducing-coral-npu-a-full-stack-platform-for-edge-ai/

Let’s start with a language

One language to find them

One language to bind them



One language to find them

One language to bind them

One language to find them

One language to bind them

Support for concurrent/
asynchronous programming

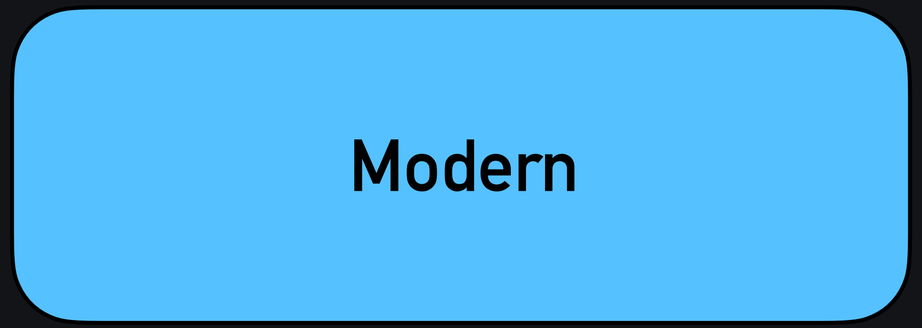
One language to find them

One language to bind them

Multiplatform

One language to find them

One language to bind them



One language to find them

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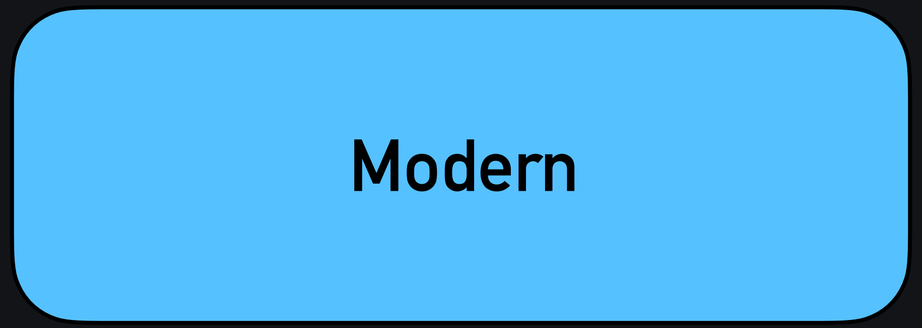

One language to find them

One language to bind them

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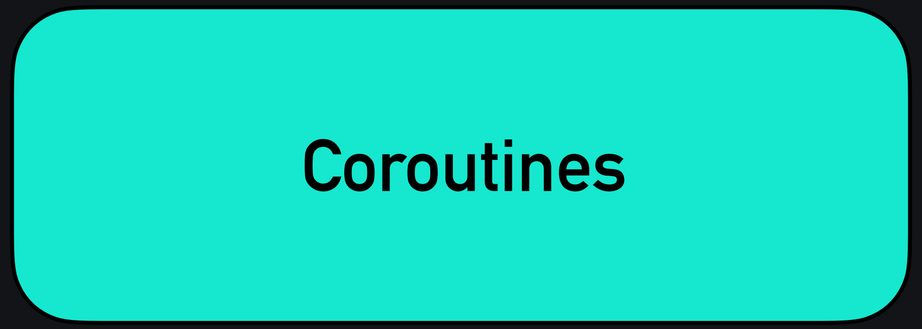


One language to find them

One language to bind them

Multiplatform

Kotlin

Kotlin is the Language for AI

https://kotlinlang.org/docs/kotlin-brand-assets.html

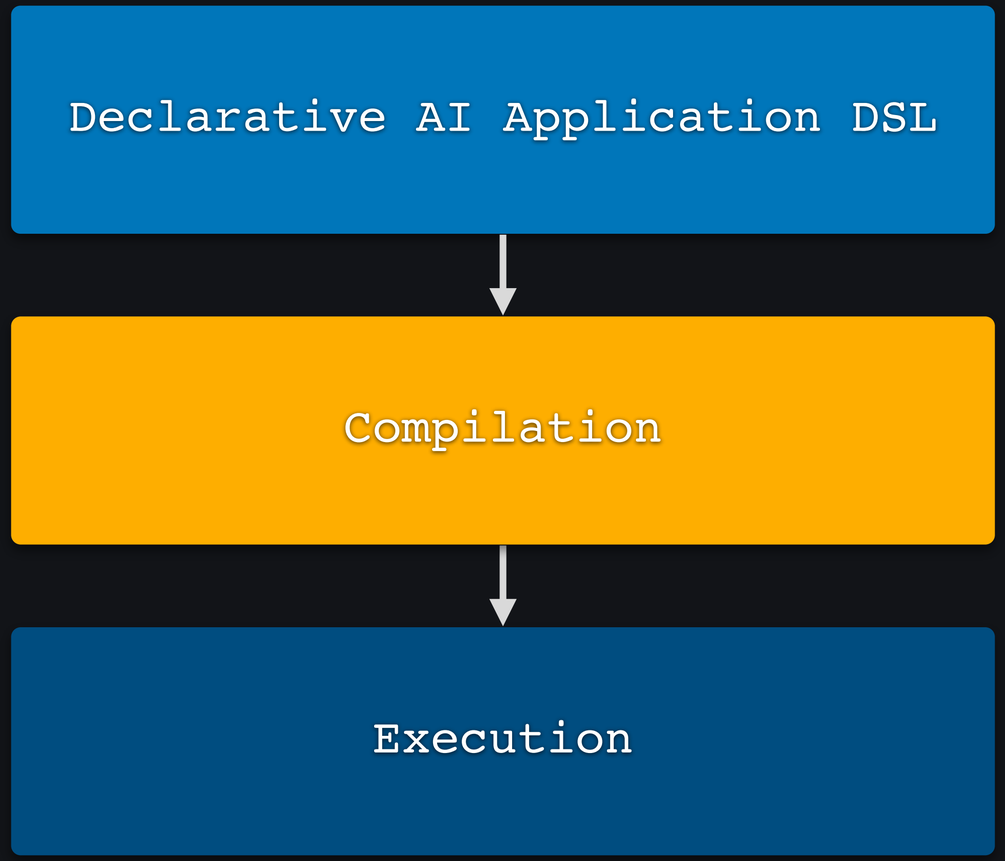
Kotlin is the Language for AI

Kotlin

- Supports DSLs for readable, domain-specific code
- Type-safe and less error-prone
- Concise, readable syntax
- Supports coroutines for asynchronous/concurrent programming
- Offers compiler plugins and tooling
- Multiplatform

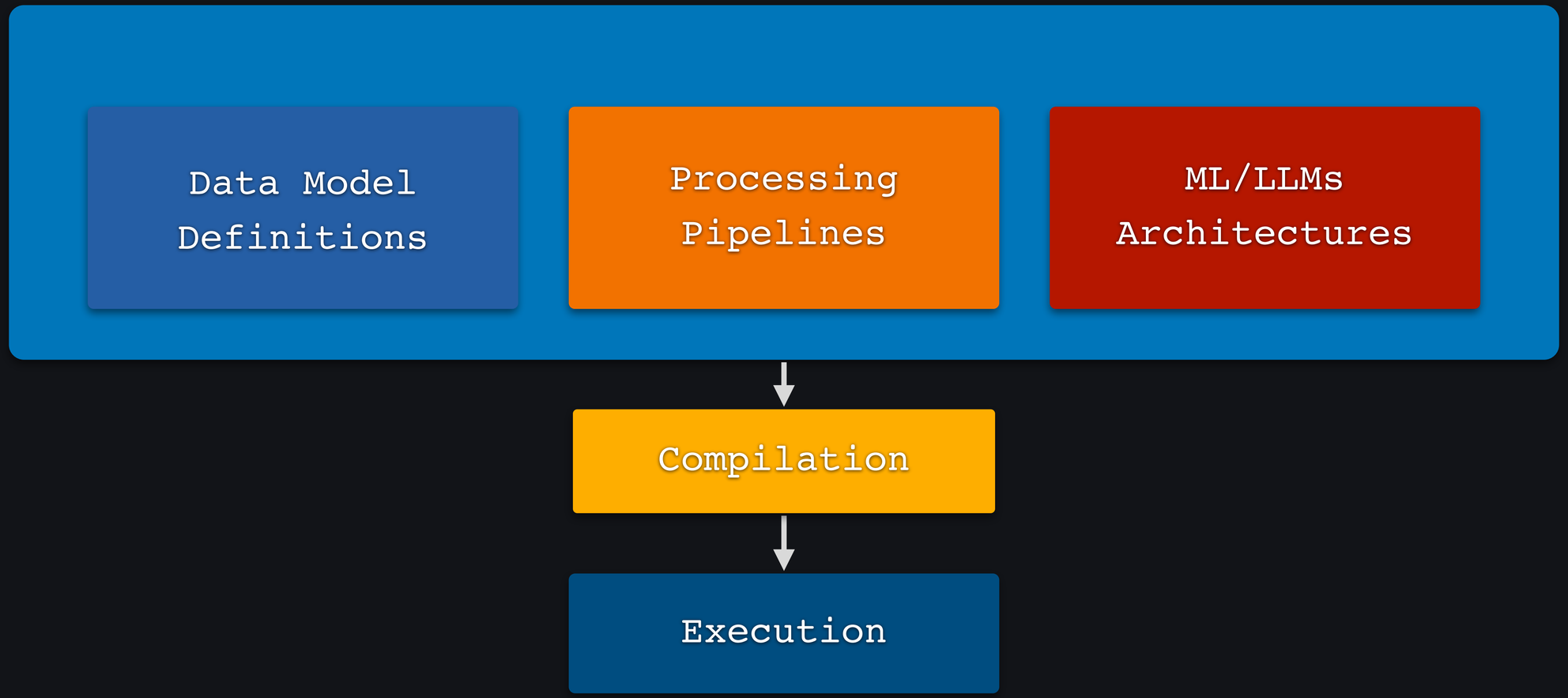
Build AI from scratch in Kotlin

Redefining on-device AI with Kotlin



Kotlin DSL

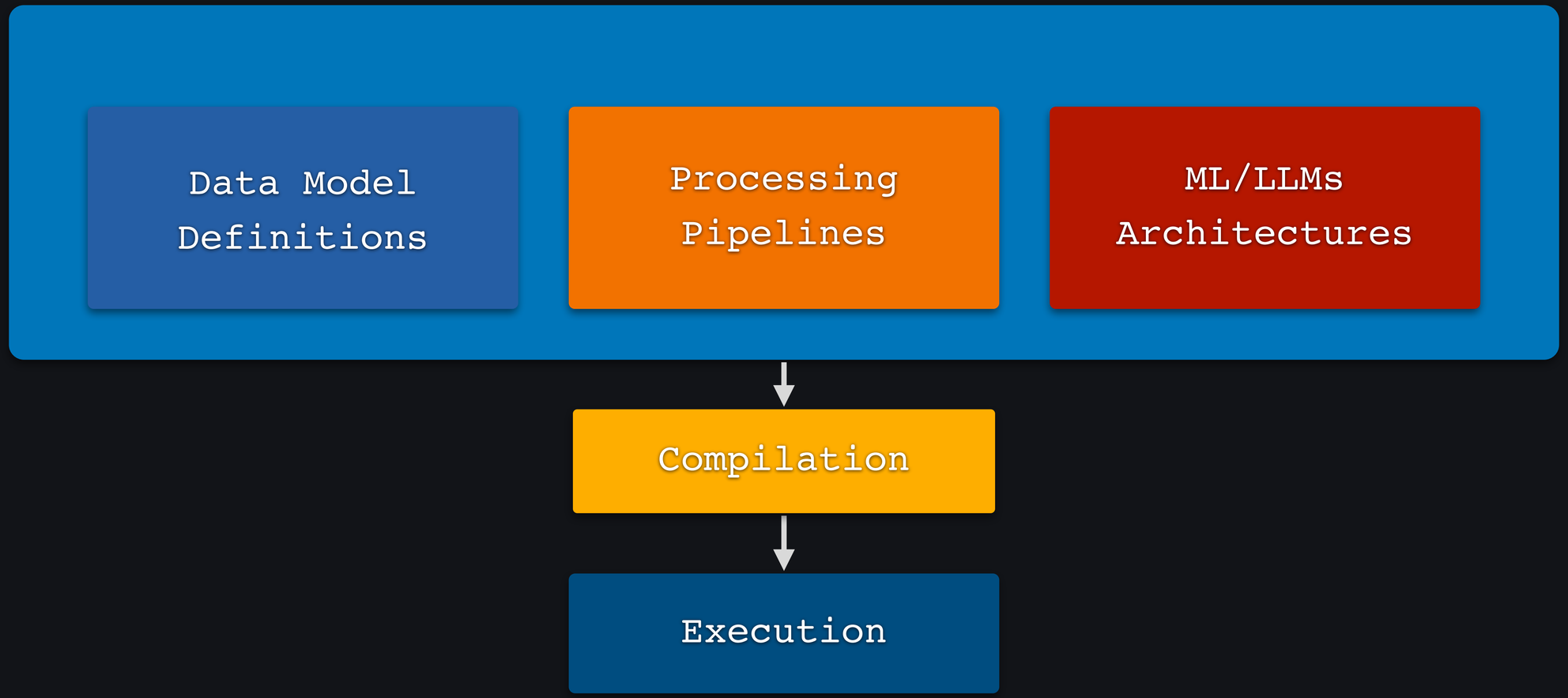
Turning an AI application into a structured Kotlin definition.



Data Model Definitions

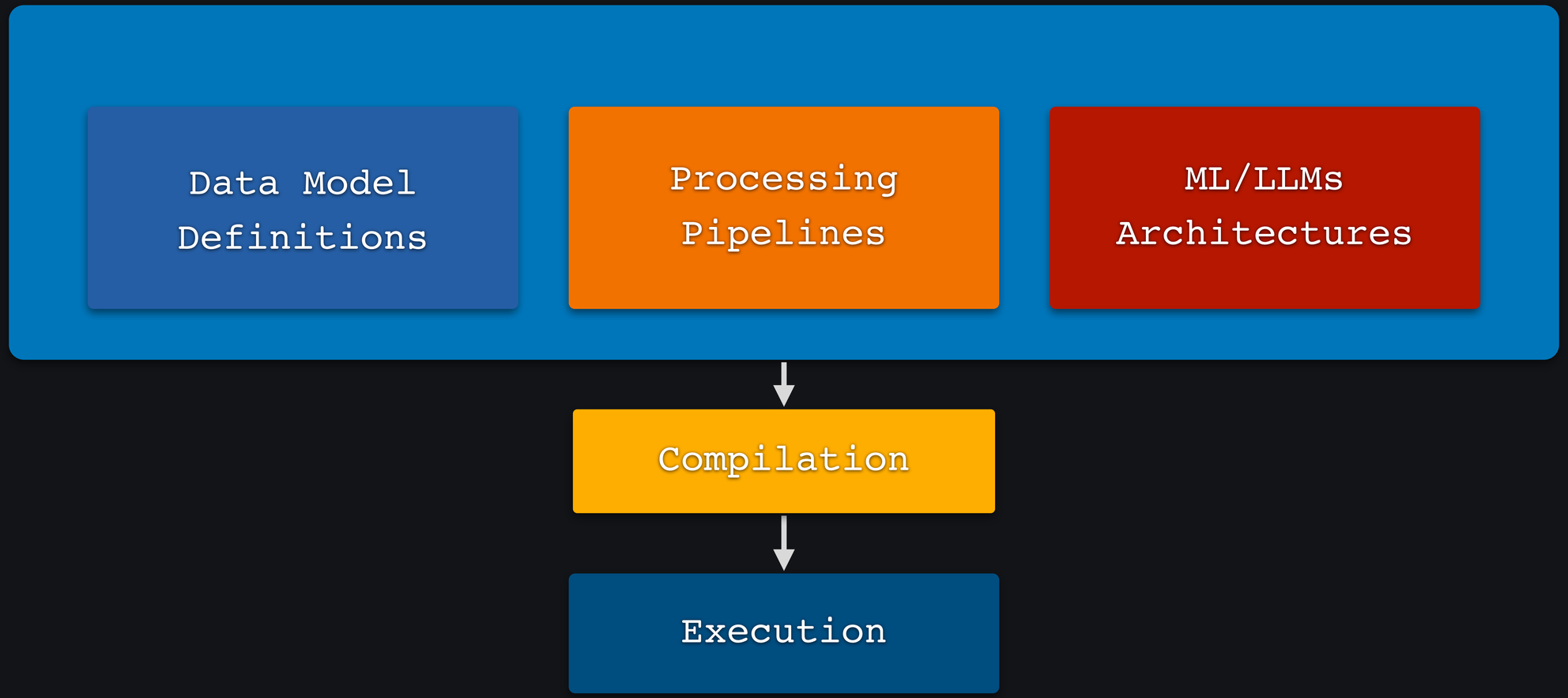
```
data {
    // RGB image
    val input = tensor<FP32, Float> {
        shape(640, 480, 3) {
            zeros()
        }
    }

    // matrix
    val mask = tensor<INT8, Int> {
        shape(10, 10) {
            zeros()
        }
    }
}
```

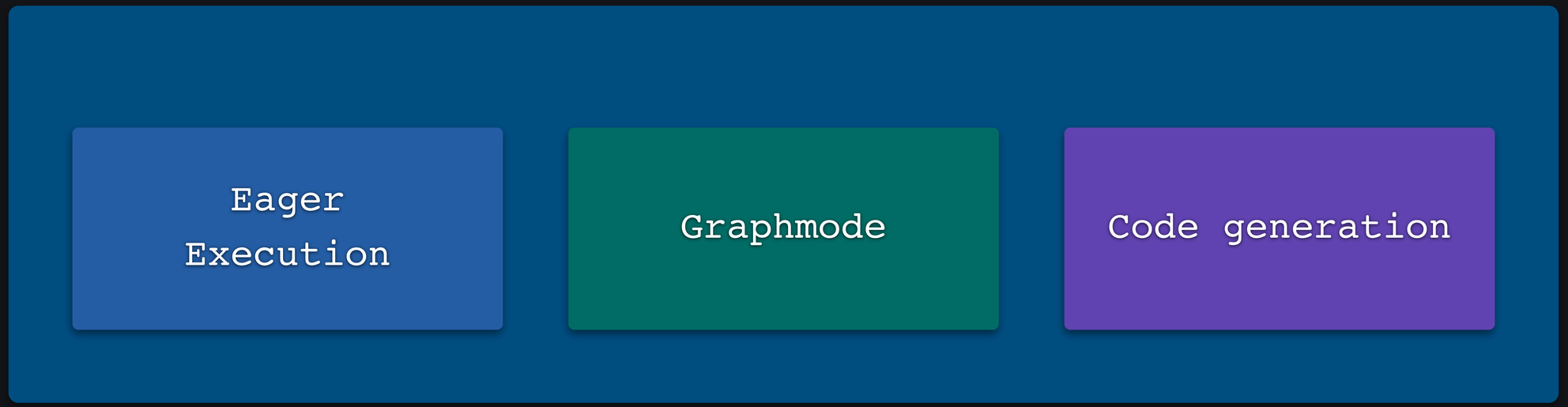
Kotlin DSL


Data Model
Definitions
Processing
Pipelines
ML/LLMs
Architectures
Compilation
Execution

Compilation

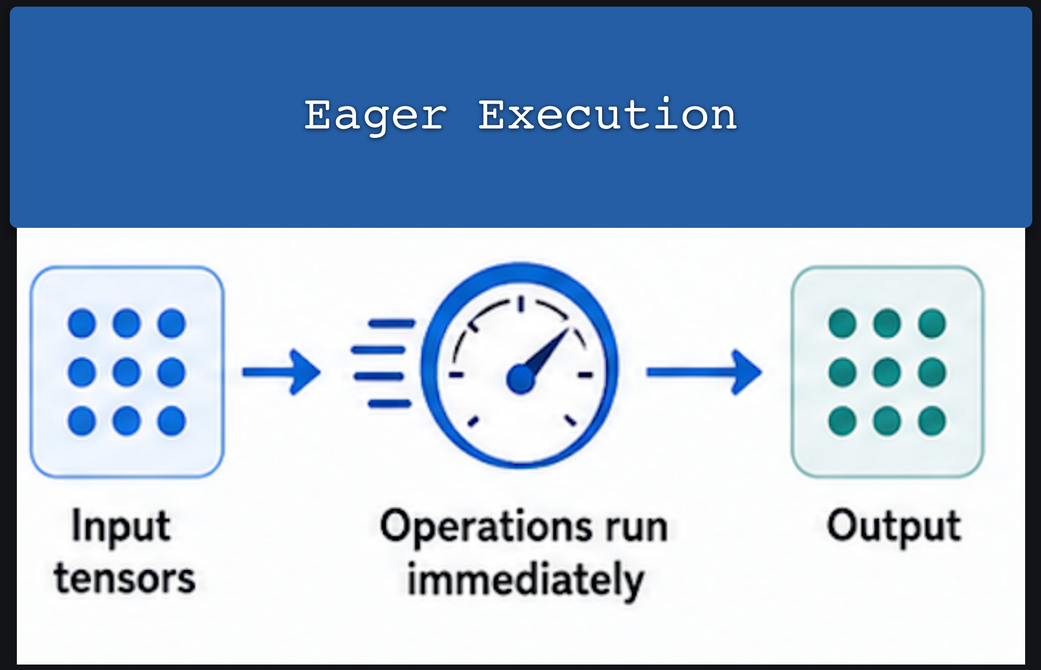
- Kotlin KSP
- LLVM.org compiler tools
  - MLIR
- IREE.org


Data Model Definitions
Processing Pipelines
ML/LLMs Architectures
Compilation
Execution


Eager
Execution
Graphmode
Code generation

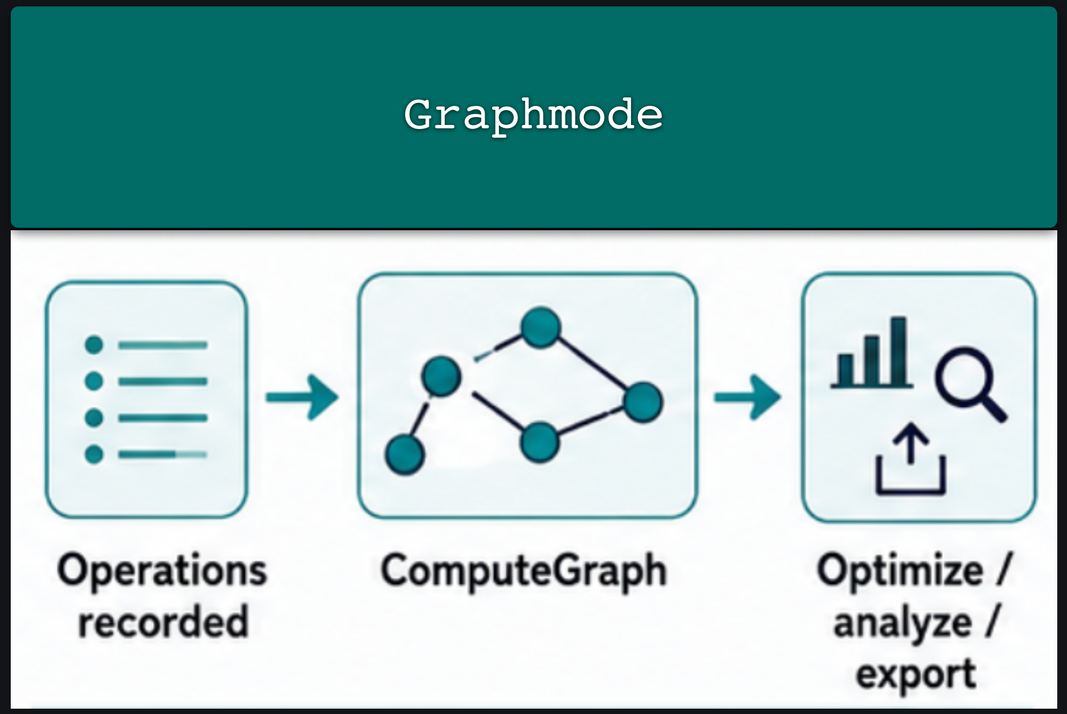
Eager Execution

DirectCpuExecutionContext



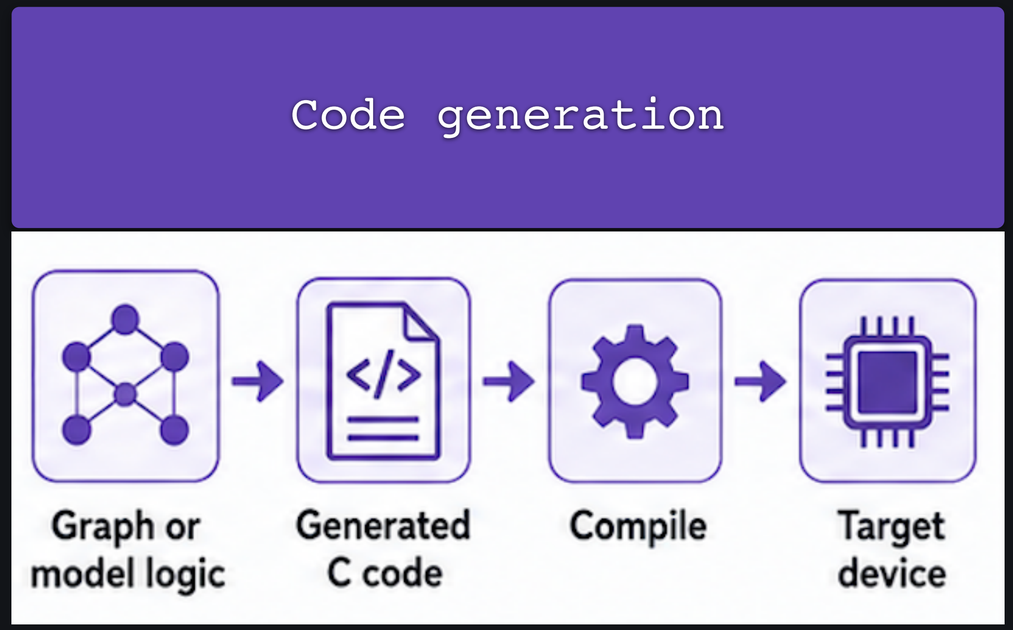
Graph mode

DefaultGraphExecutionContext



Code generation

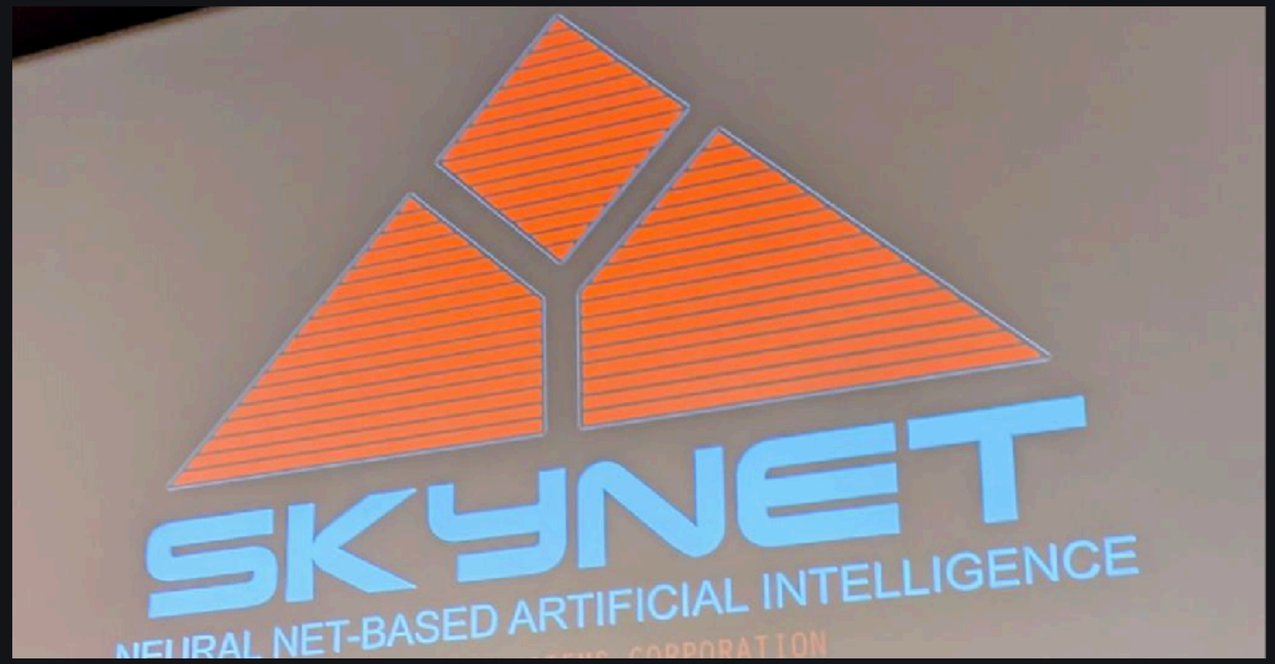
Compiled deployment path



One more thing …

SKaiNET

Naming is hard - a working title



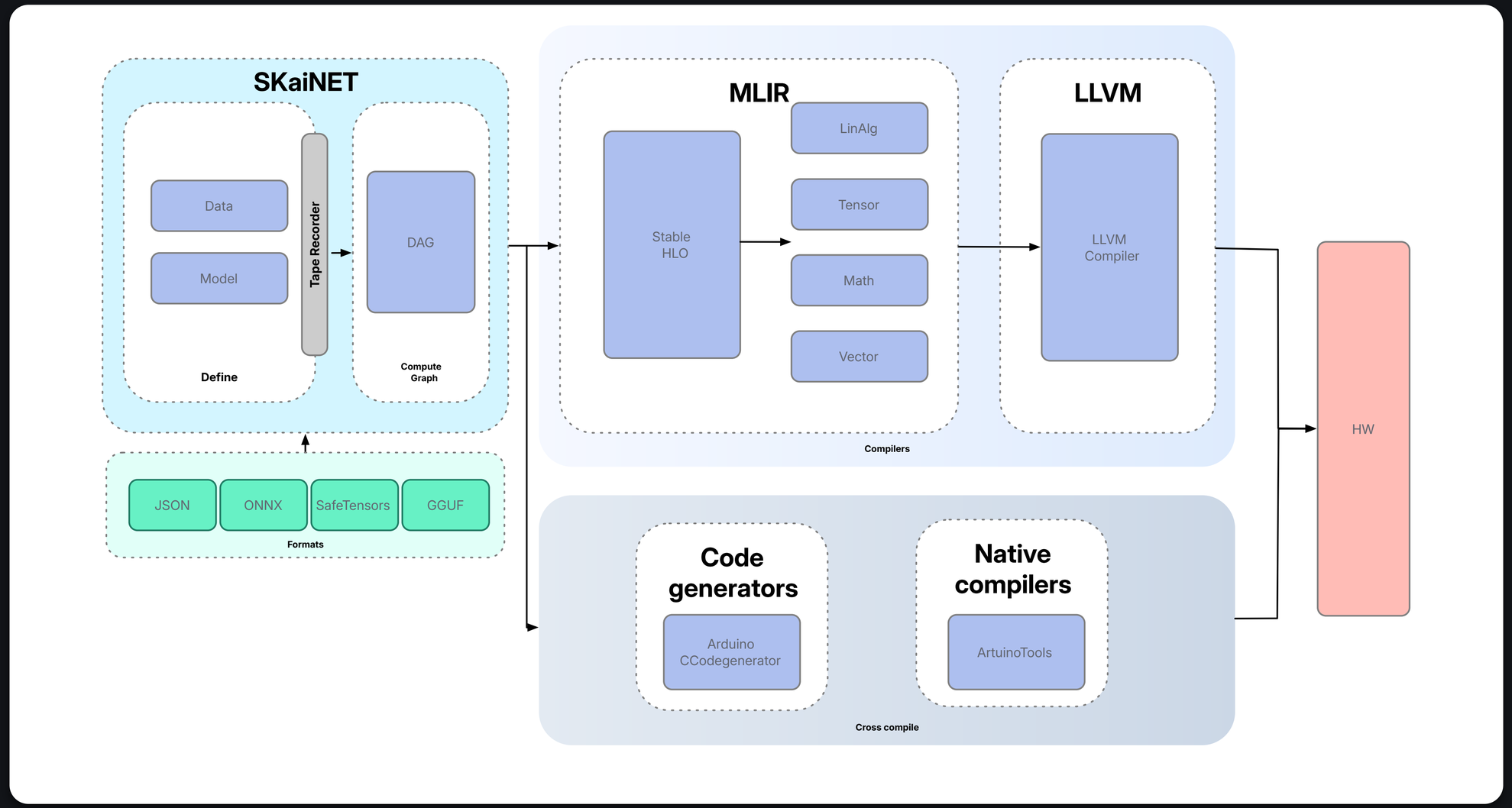
Kotlin

Yet Another ML/AI Framework ?

- On-device-first
- Platform- and hardware-agnostic
- Developer-focused
- Kotlin powered

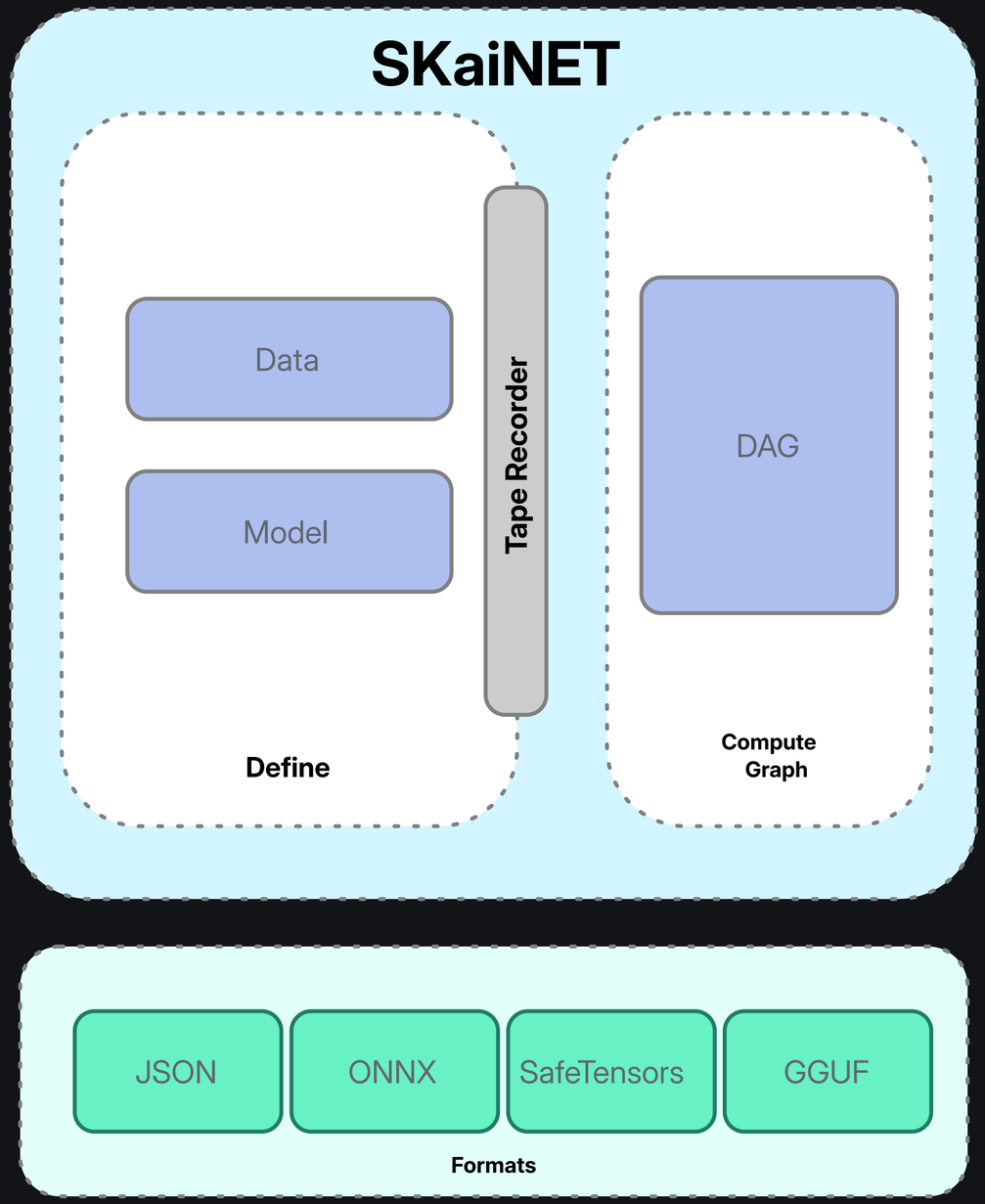
Architecture

Standing on the shoulders of giants



SKaiNET

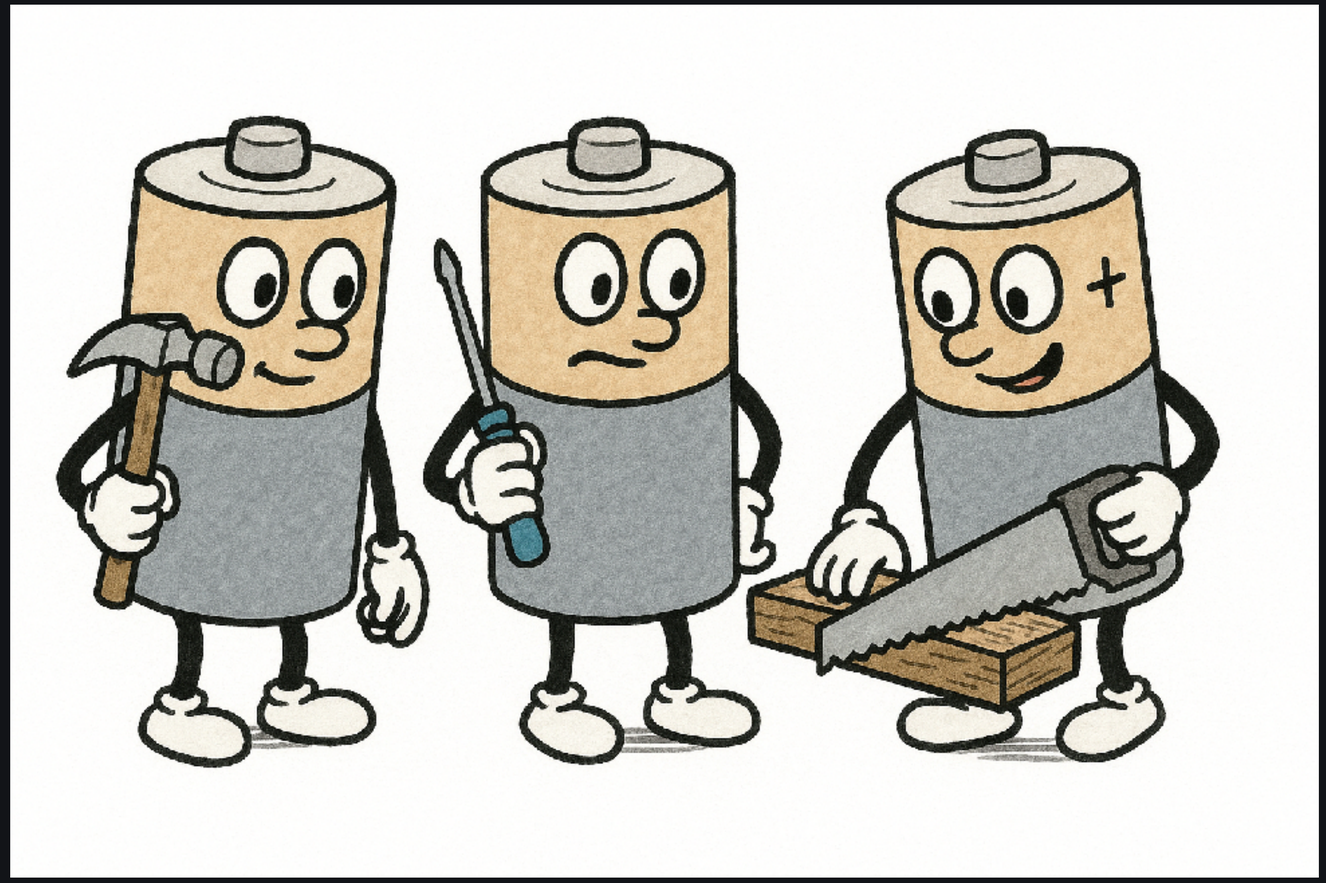
Core



SKaiNET is open source

MIT license

- Collaboration
- OSS compliance effort



Open Source project development

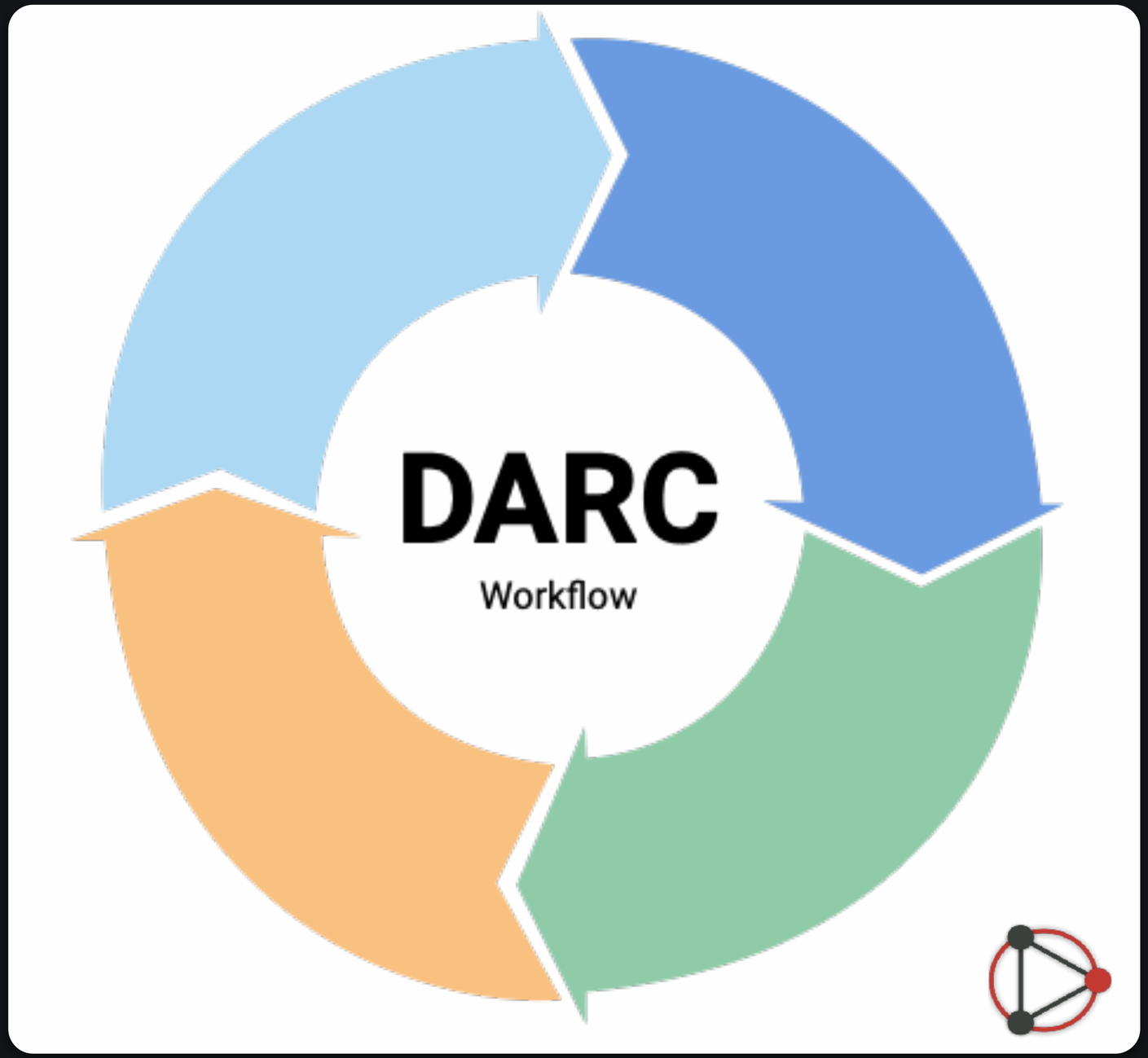
Bringing data scientist and developer together

- I can Kotlin
- I can machine learning

DARC

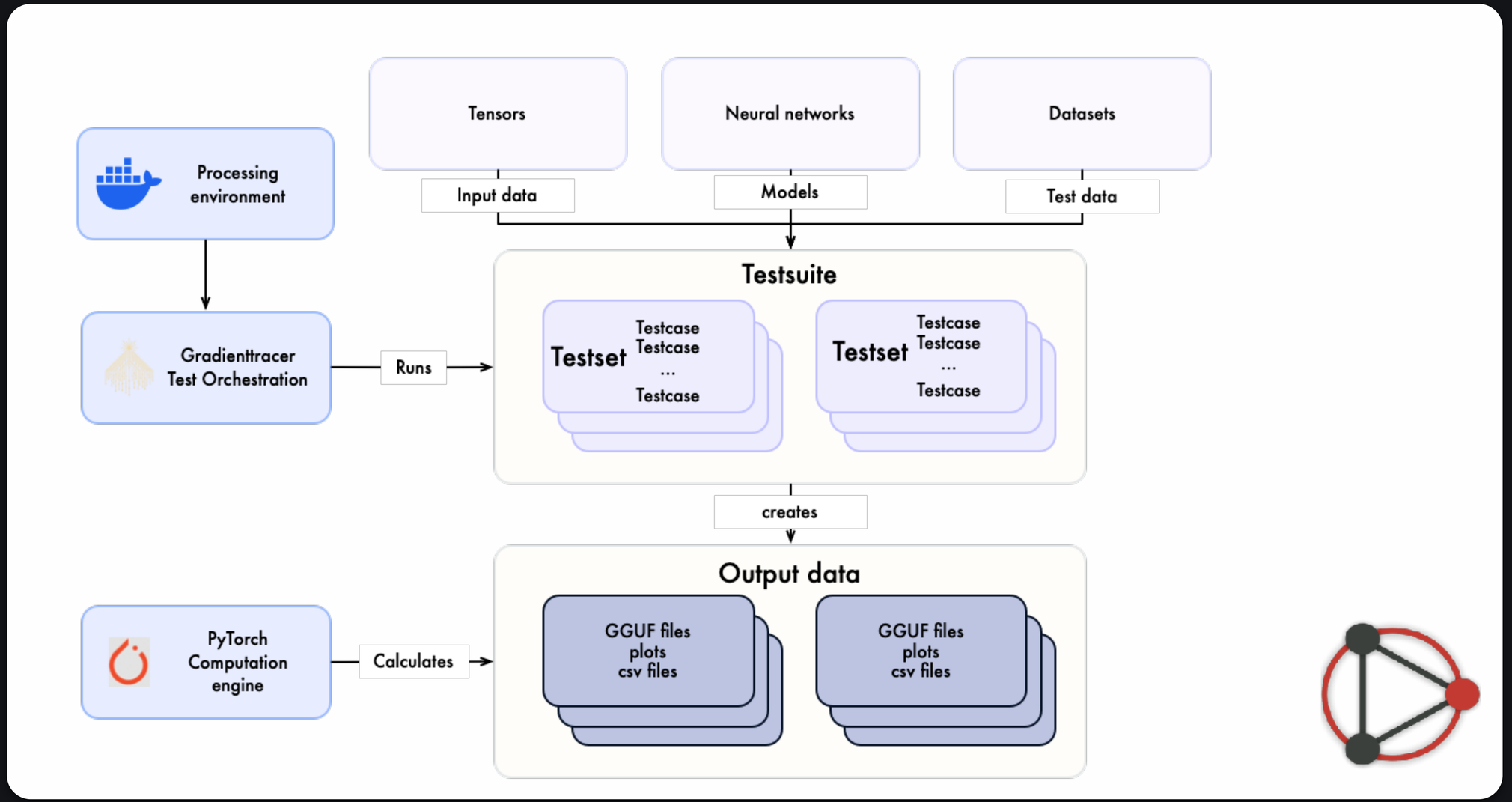
Collaboration workflow

- Document
- Assess
- Research
- Code

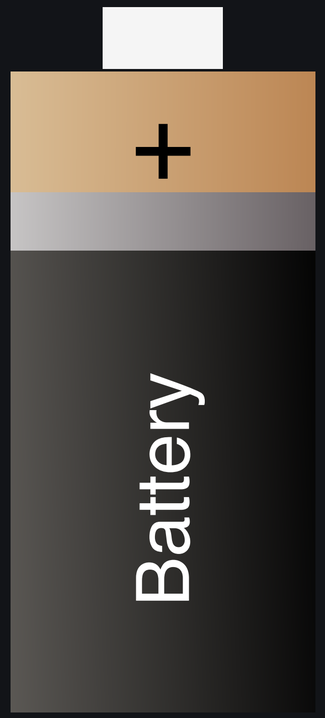


Validation

Create Ground Truth



Demo


+
Battery

https://harakal.de

Thank You, and Don’t Forget to Vote